

# Environmental Chemistry Theory

**CY1010**



भारतीय प्रौद्योगिकी संस्थान हैदराबाद  
Indian Institute of Technology Hyderabad

Aug-2026 to Oct 2026

# Course Content

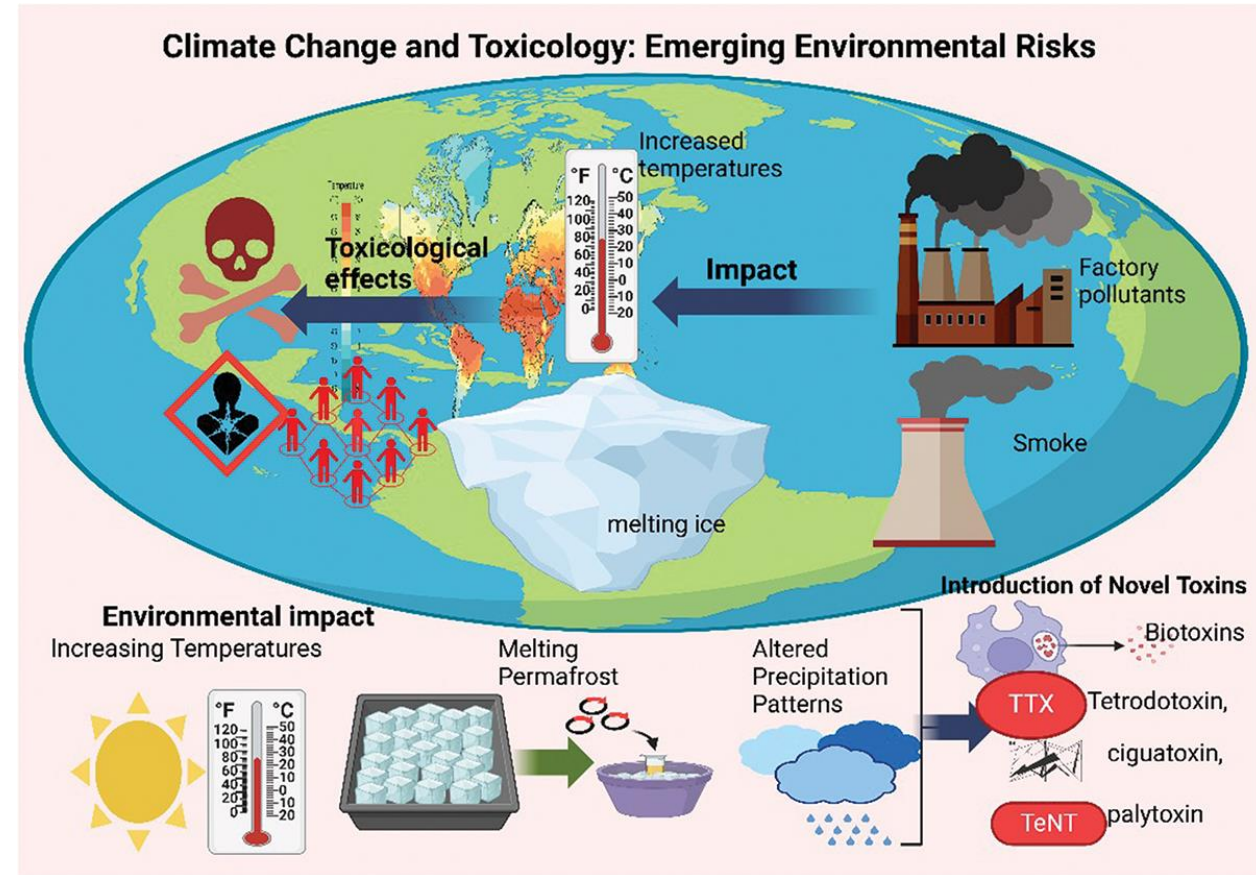
Deeper  
analysis of  
atmospheric  
pollution

Chemistry of CO, NO<sub>x</sub>,  
VOCs, SO<sub>2</sub>, Industrial  
smog, photochemical  
smog

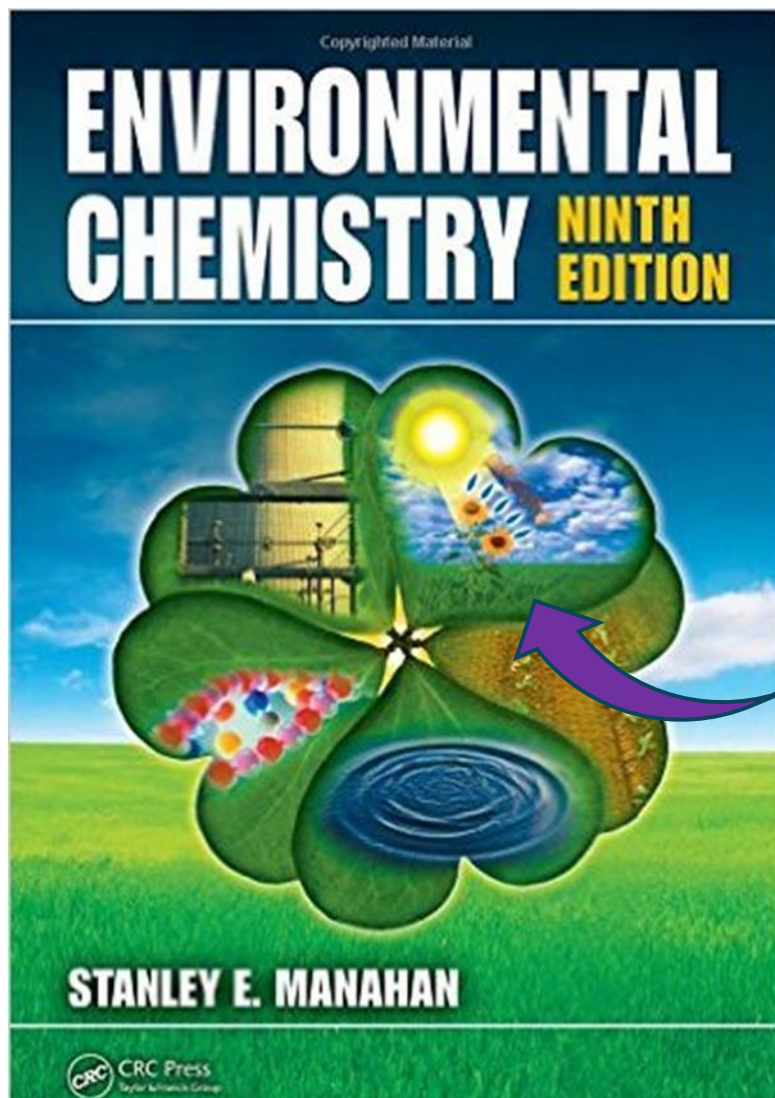
Ozone depletion

Production

Catalytic  
destruction



# Reference books

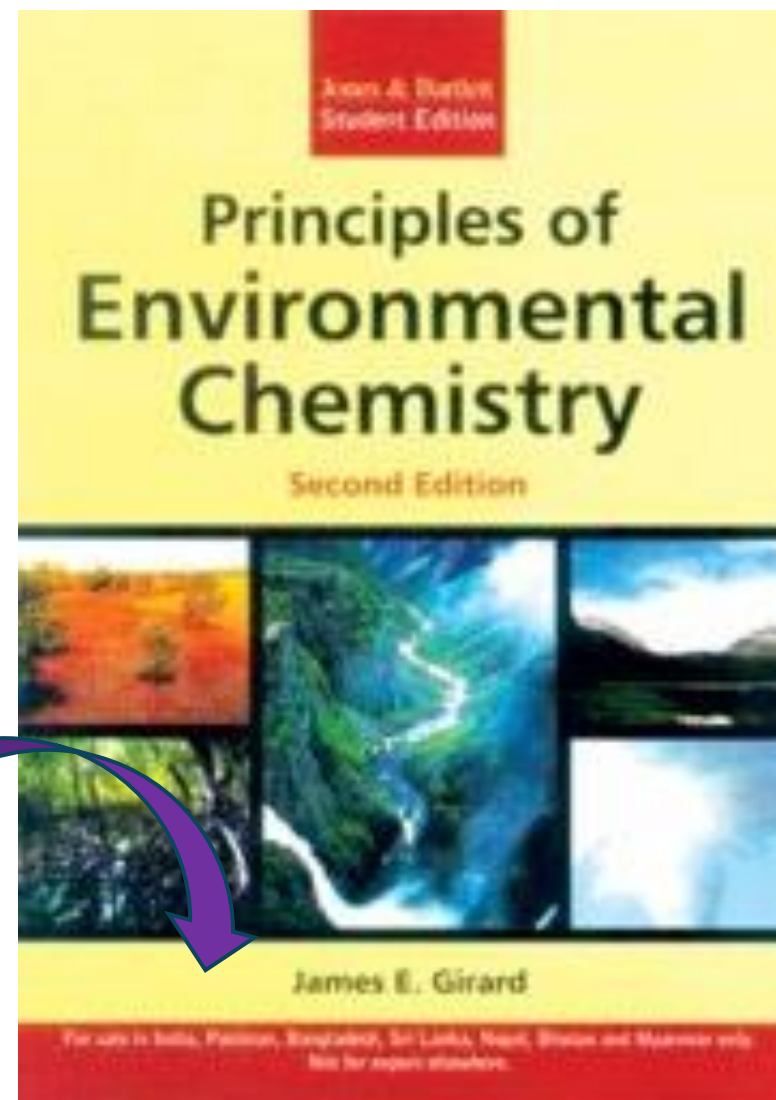


## ENVIRONMENTAL CHEMISTRY

NINTH EDITION

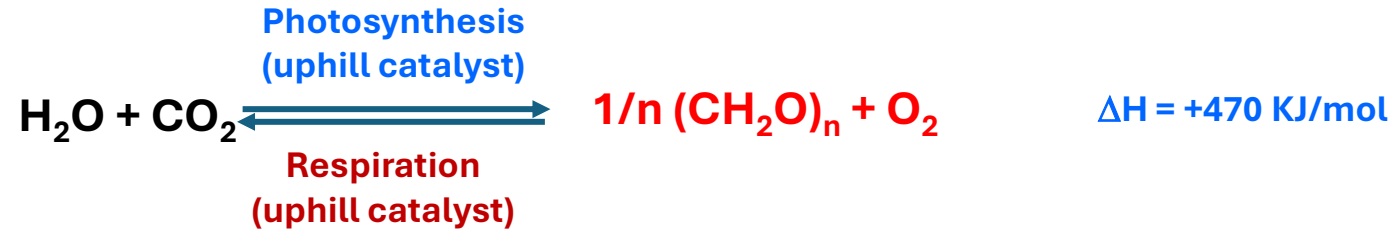
STANLEY E. MANAHAN

Principles of  
Environmental  
Chemistry  
By James E. Girard



# Photosynthesis

*The most IMPORTANT chemical reaction for life*



## Two Nobel Prizes:

**1992:** Rudolph A. Marcus



**“for contributions to the theory of electron transfer reactions in chemical systems”**

## 1988 Nobel Prize in Chemistry



Johann Deisenhofer  
Prize share: 1/3



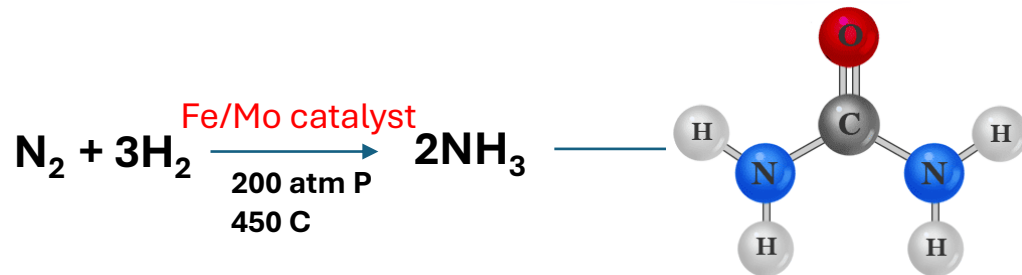
Robert Huber  
Prize share: 1/3



Hartmut Michel  
Prize share: 1/3

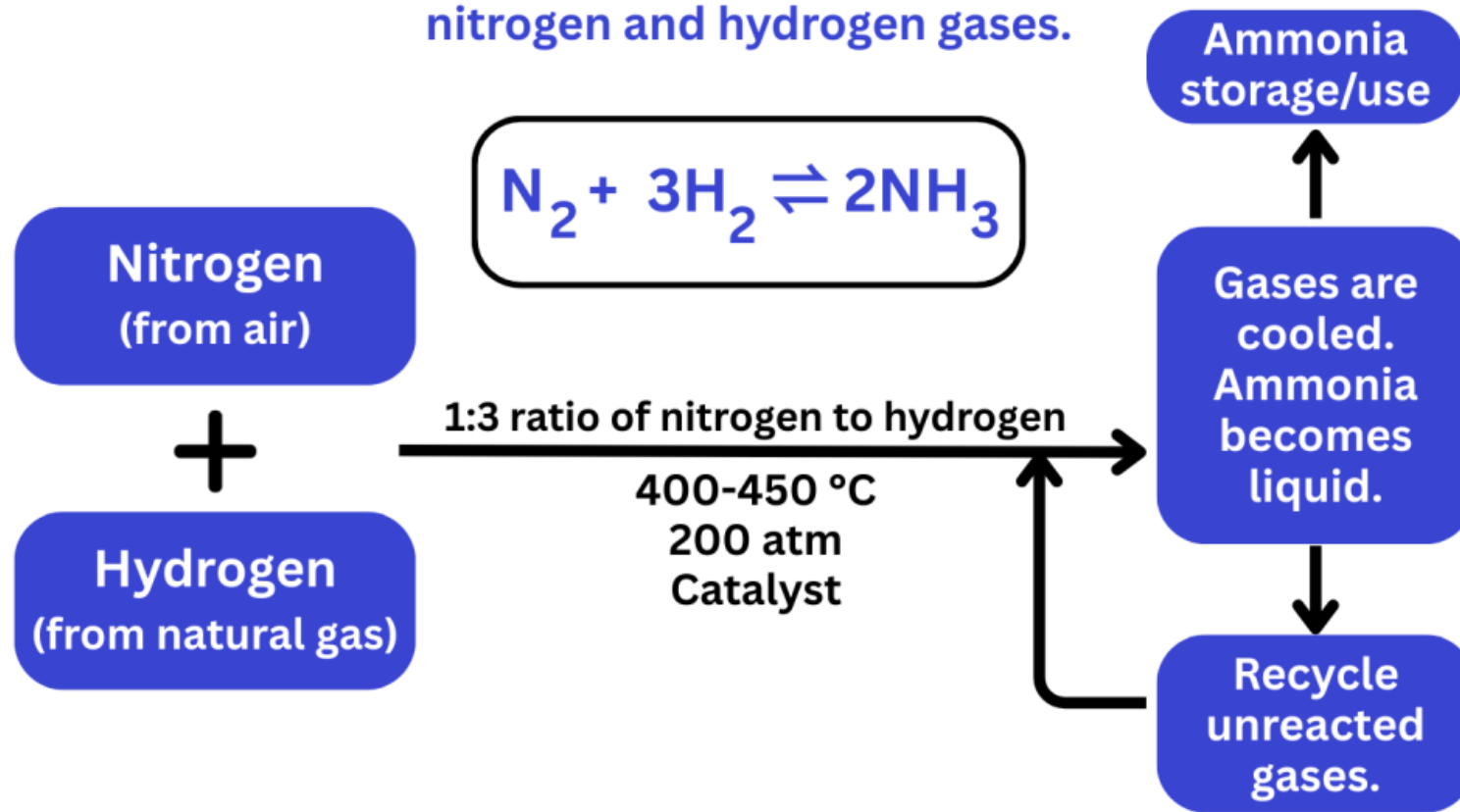
**"for the determination of the three-dimensional structure of a photosynthetic reaction centre"**

**(Bacterial Photosynthesis Process)**



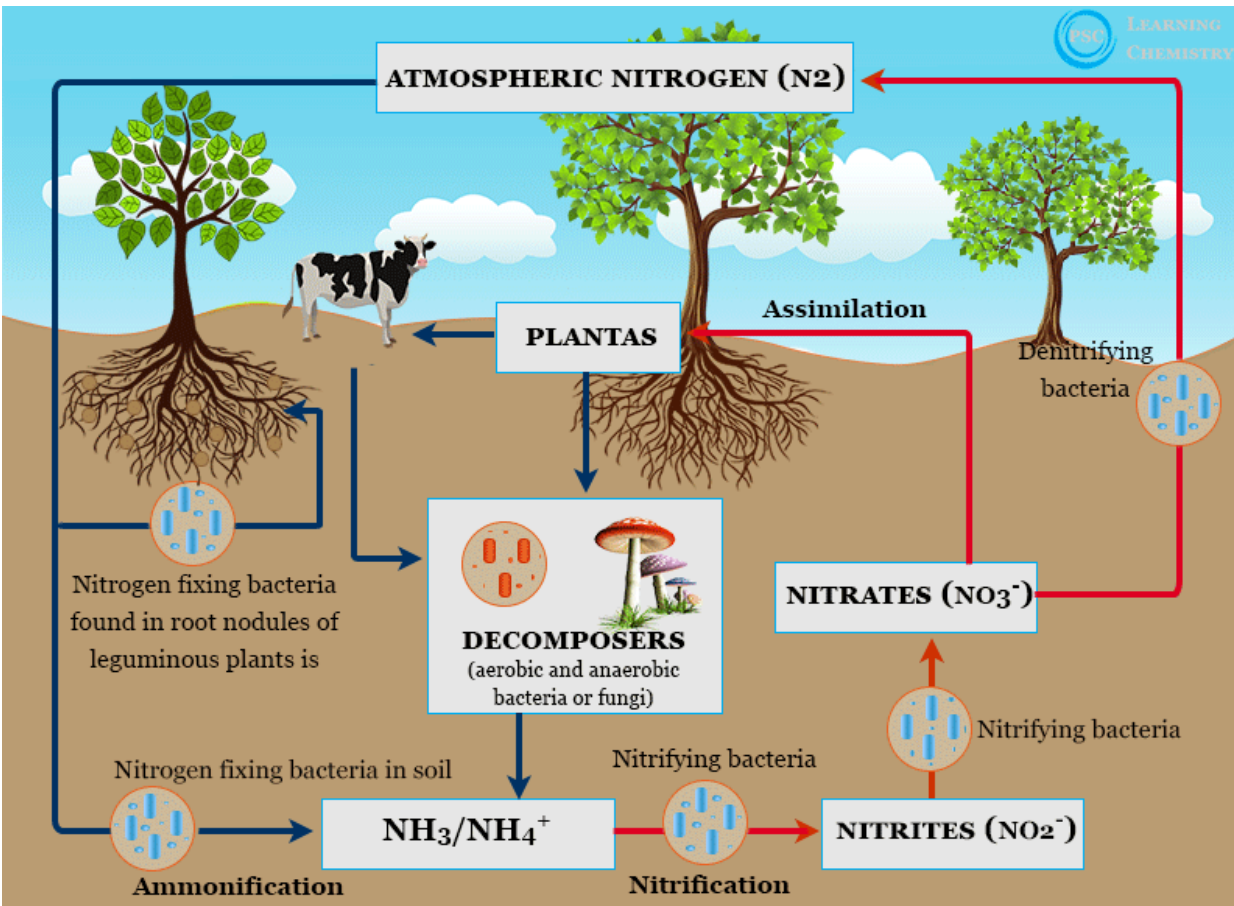
## Haber-Bosch Process

The Haber-Bosch process synthesizes ammonia from nitrogen and hydrogen gases.



# Nitrogen Fixation

## Nitrogen-fixing Leguminous Plant: *Pisum sativum*



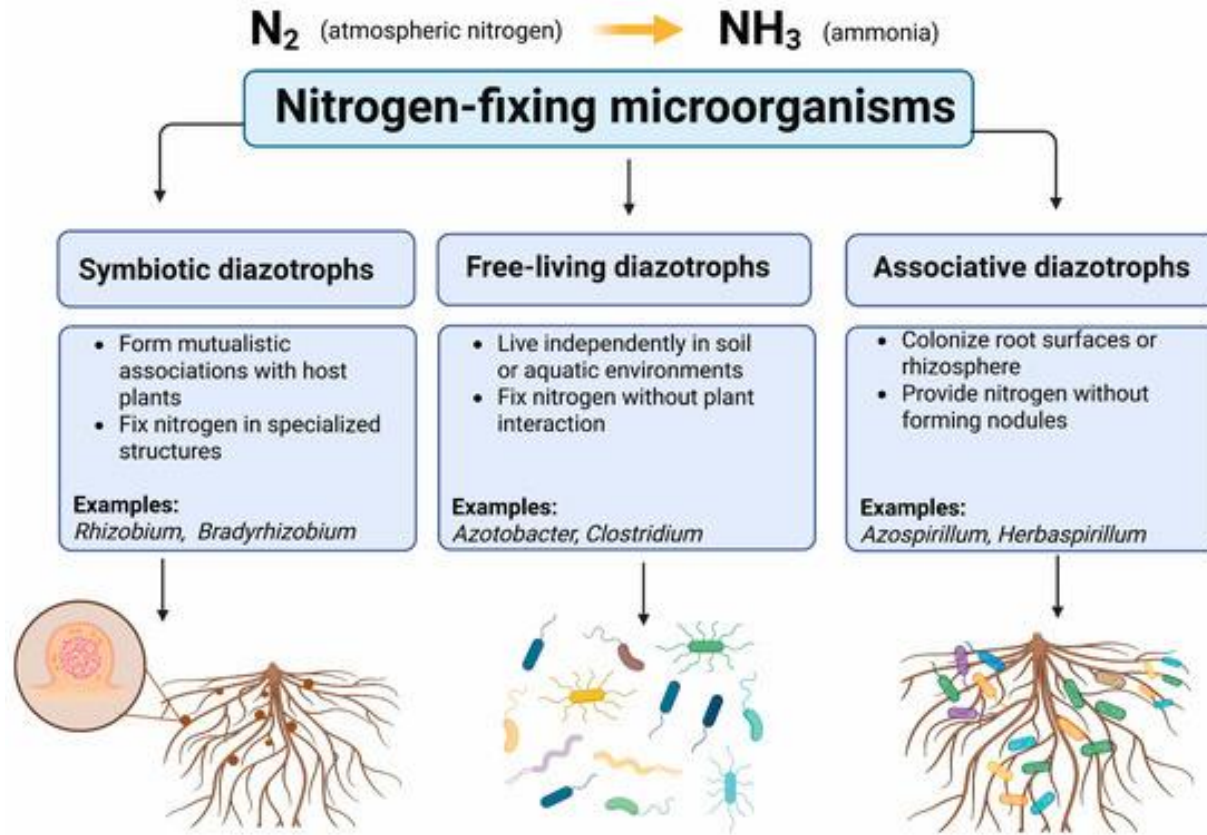
Highly energy-intensive process consuming around 2% of global energy output each year (steam methane reforming accounts for over 80% of the energy required) and producing as a result about 500 million tonnes of carbon dioxide (about 1.8% of global carbon dioxide emissions)

**Haber** and **Bosch** were later **awarded Nobel Prizes**, in **1918** and **1931** respectively, for their work in overcoming the chemical and engineering problems of large-scale, continuous-flow, high-pressure technology.

Synthetic ammonia from the Haber process was used for the production of nitric acid, a precursor to the nitrates used in explosives.

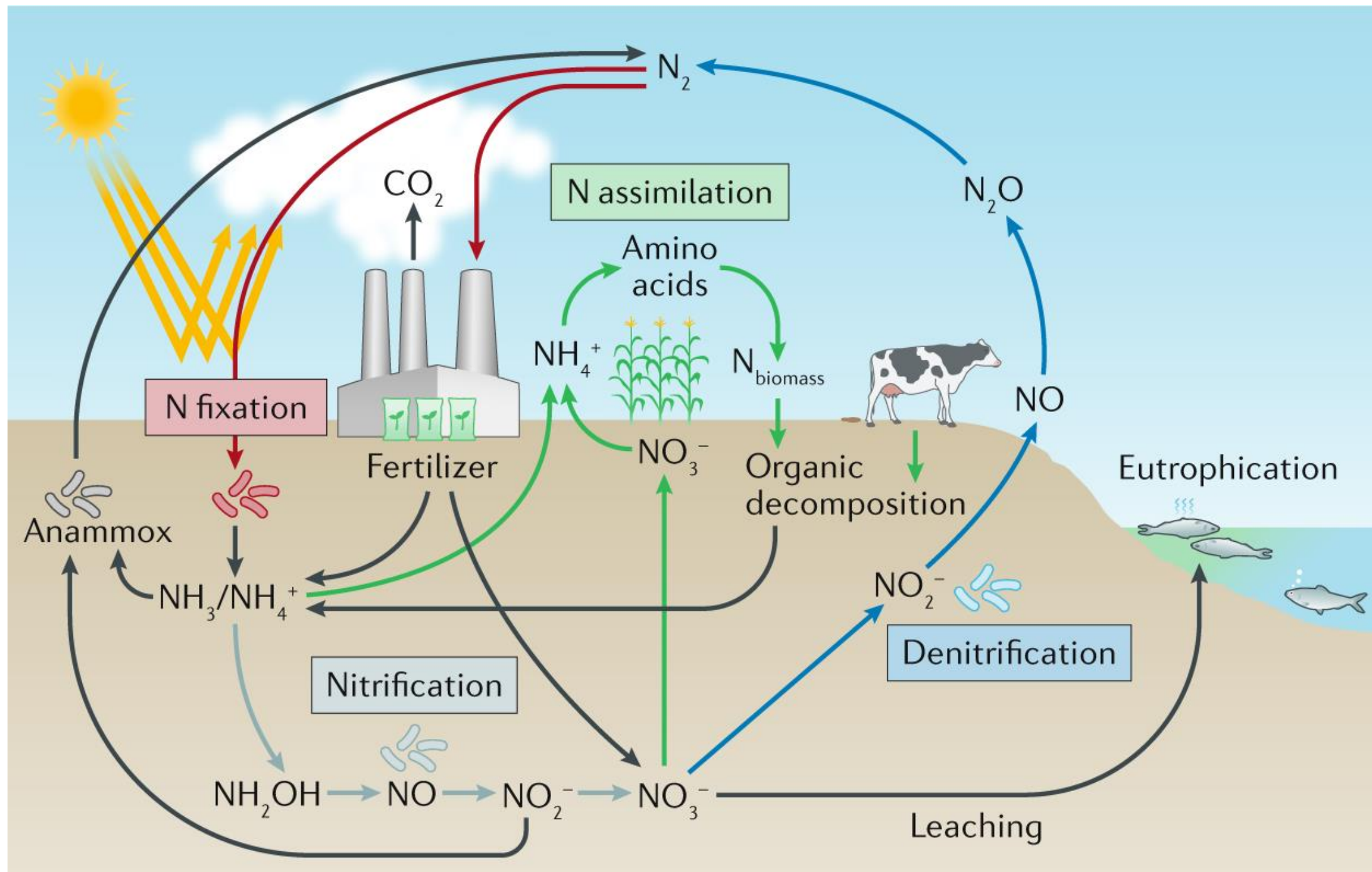
**Nitric Oxide (NO)?**

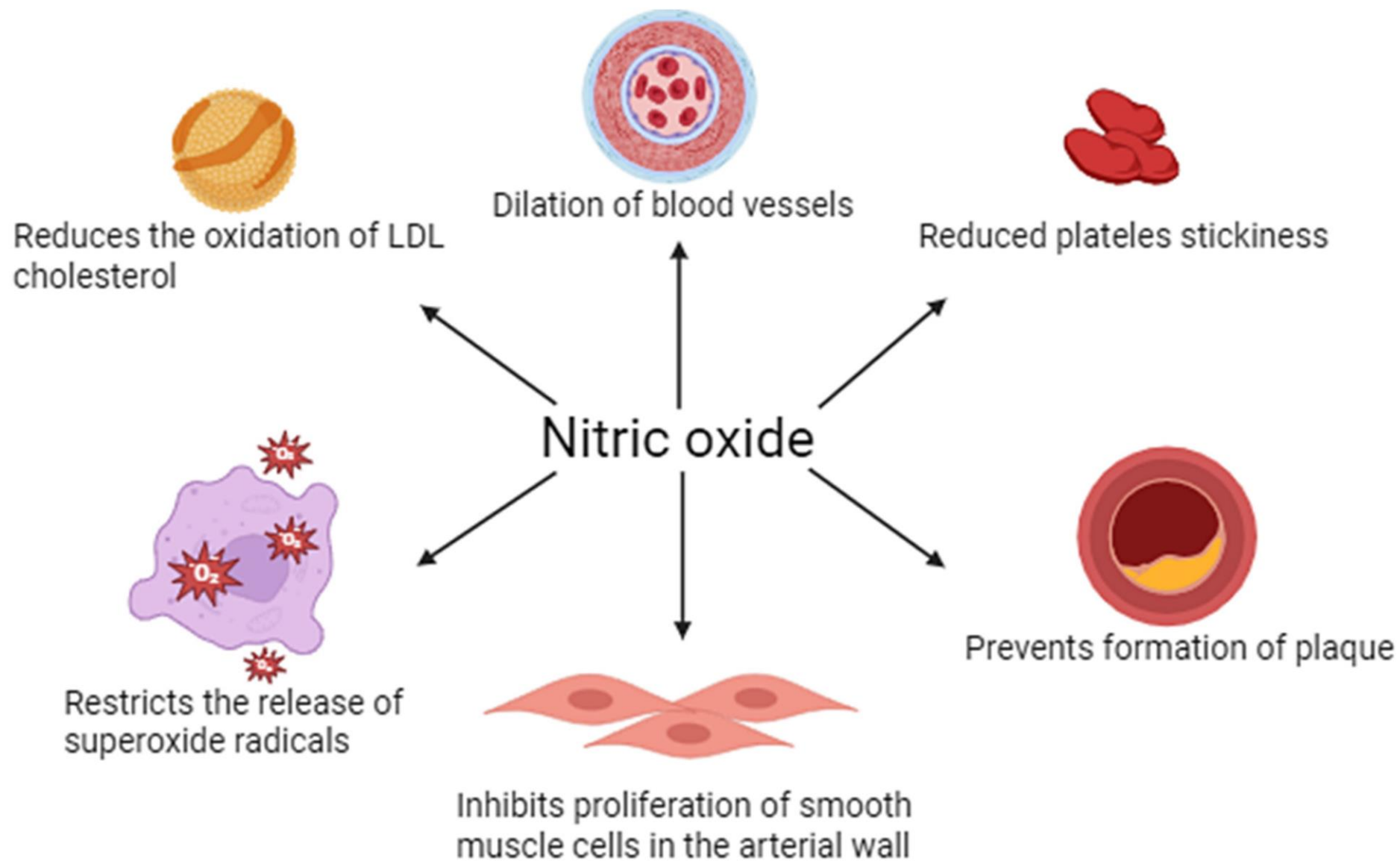
The roots of leguminous plants have nodules that contain  $N_2$ -fixing bacteria



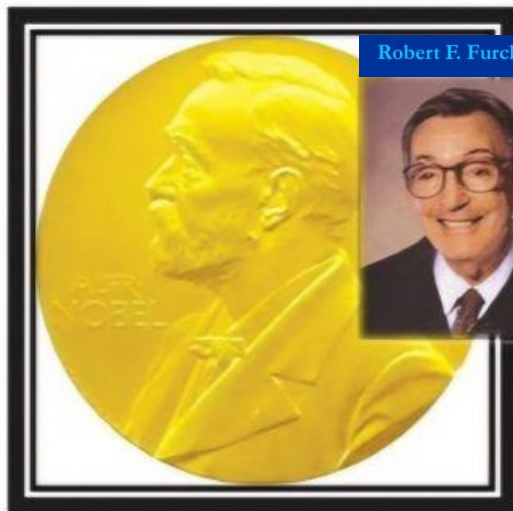
**Nitrogenase:** key enzyme for biological nitrogen fixation.

**The molybdenum-iron (MoFe) protein** is the large, catalytic component (component 1) of the nitrogenase enzyme complex. It is a  $\alpha_2\beta_2$  heterotetramer that binds metal cofactors (P-clusters and FeMo-cofactors), receiving electrons from the Fe protein to reduce atmospheric nitrogen ( $N_2$ ) into usable ammonia ( $NH_3$ ).





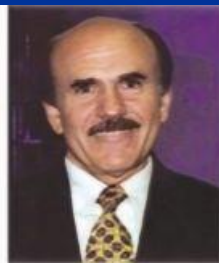
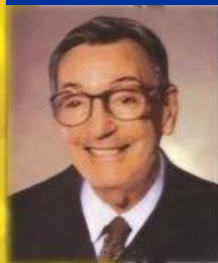
# 1998 Nobel Prize



Robert F. Furchgott

Louis J. Ignarro

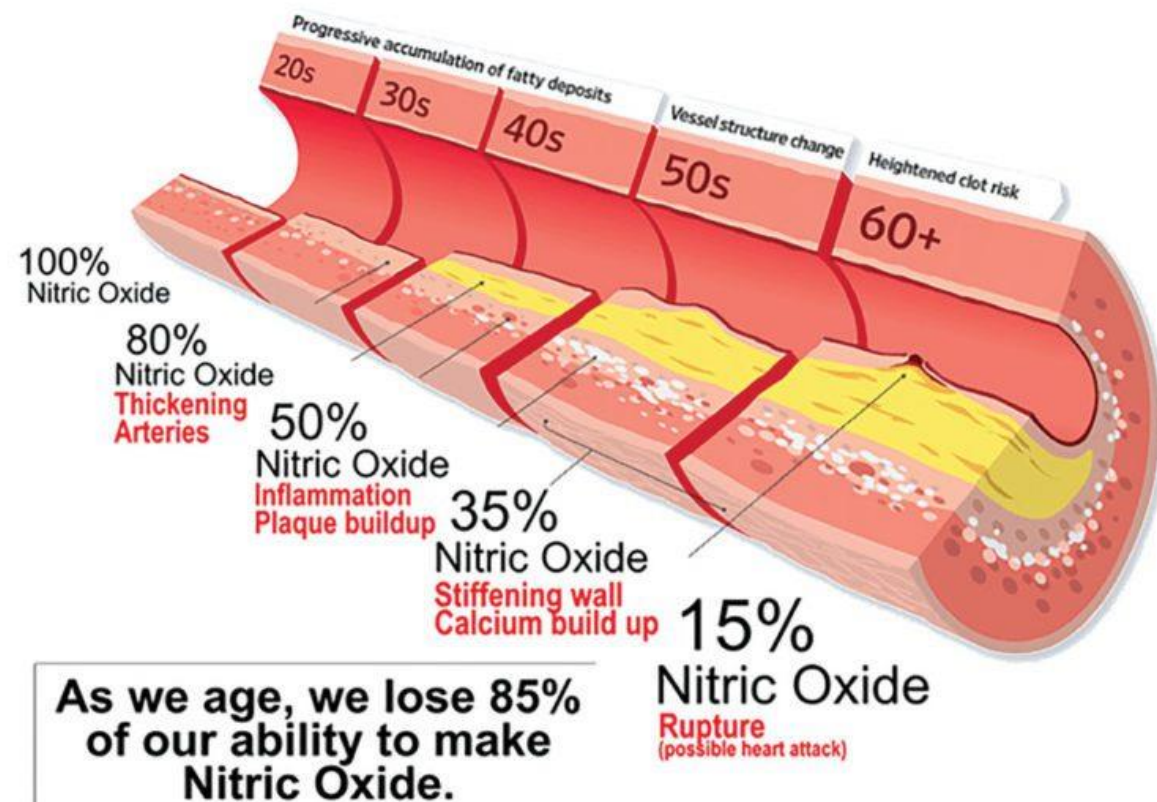
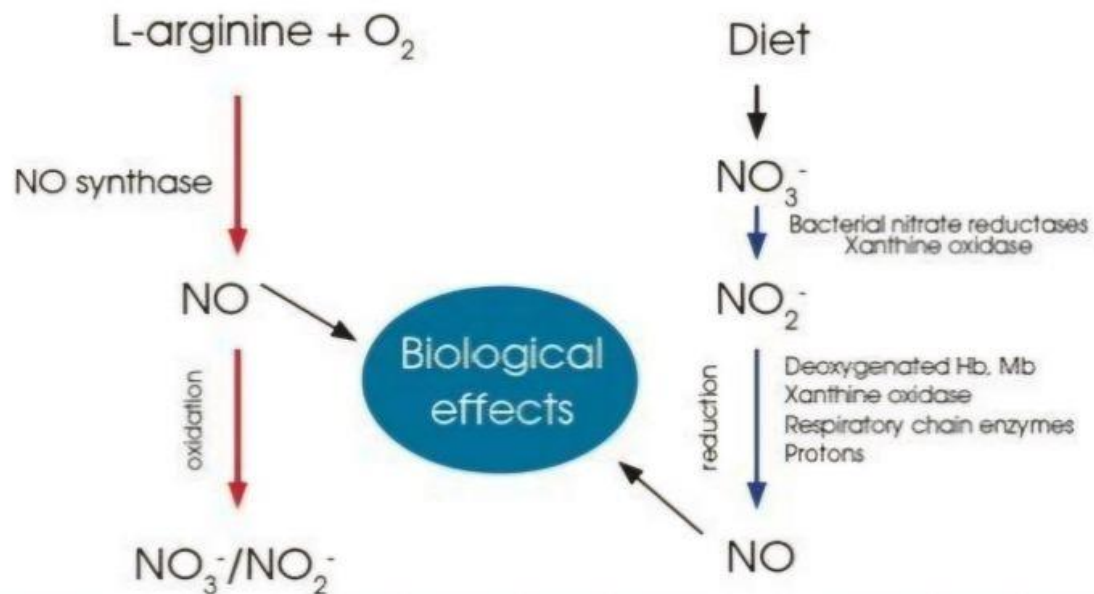
Ferid Murad



"FOR THEIR DISCOVERIES  
CONCERNING NITRIC OXIDE AS A  
SIGNALING MOLECULE IN THE  
CARDIOVASCULAR SYSTEM"

The L-arginine-nitric oxide pathway

The nitrate-nitrite-nitric oxide pathway



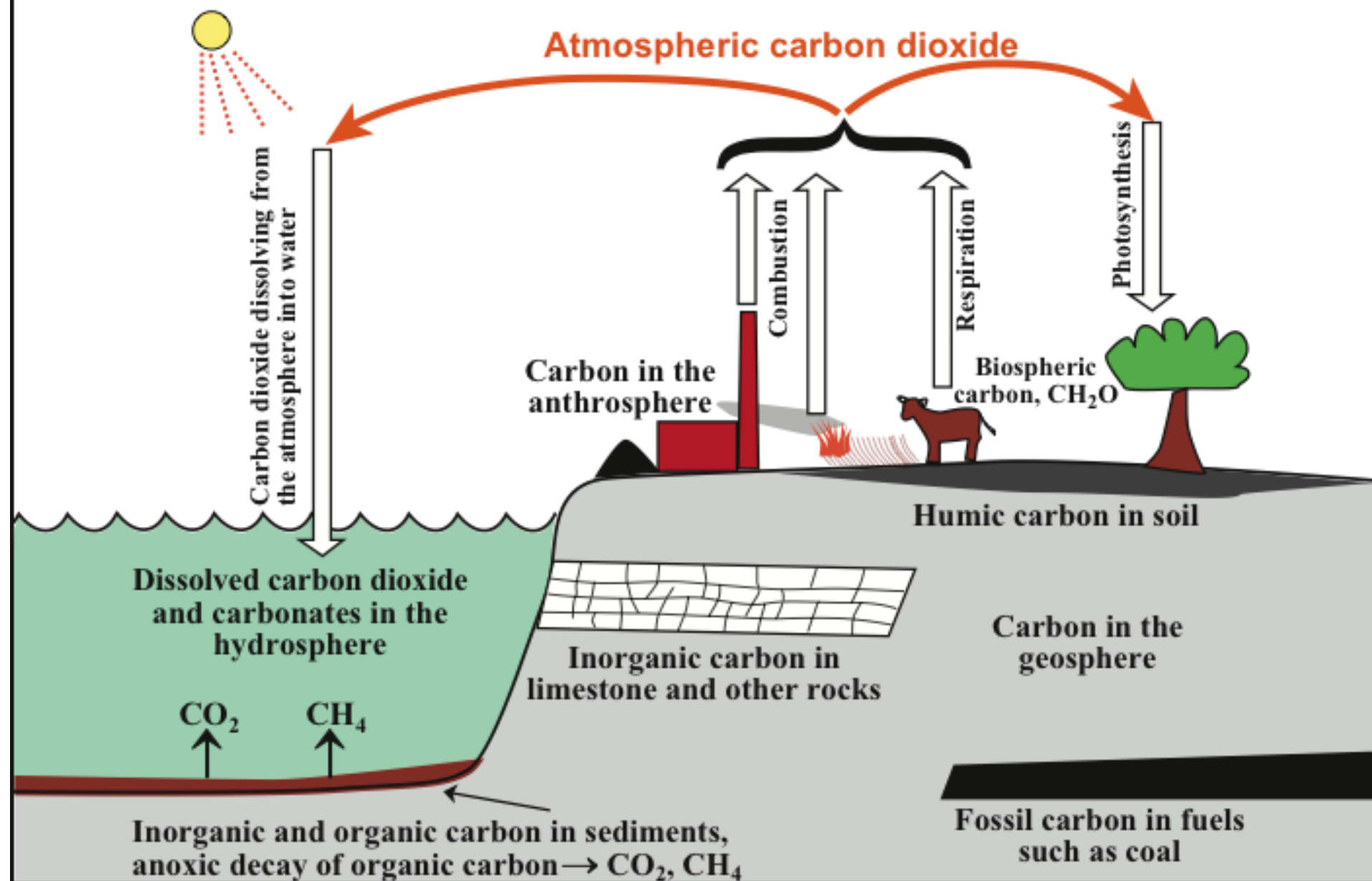
# The Five Spheres of the Environment



**Matter and Energy are Exchanged  
Largely through Biogeochemical Cycles**

# Biogeochemical Cycles Are Important in Environmental Chemistry

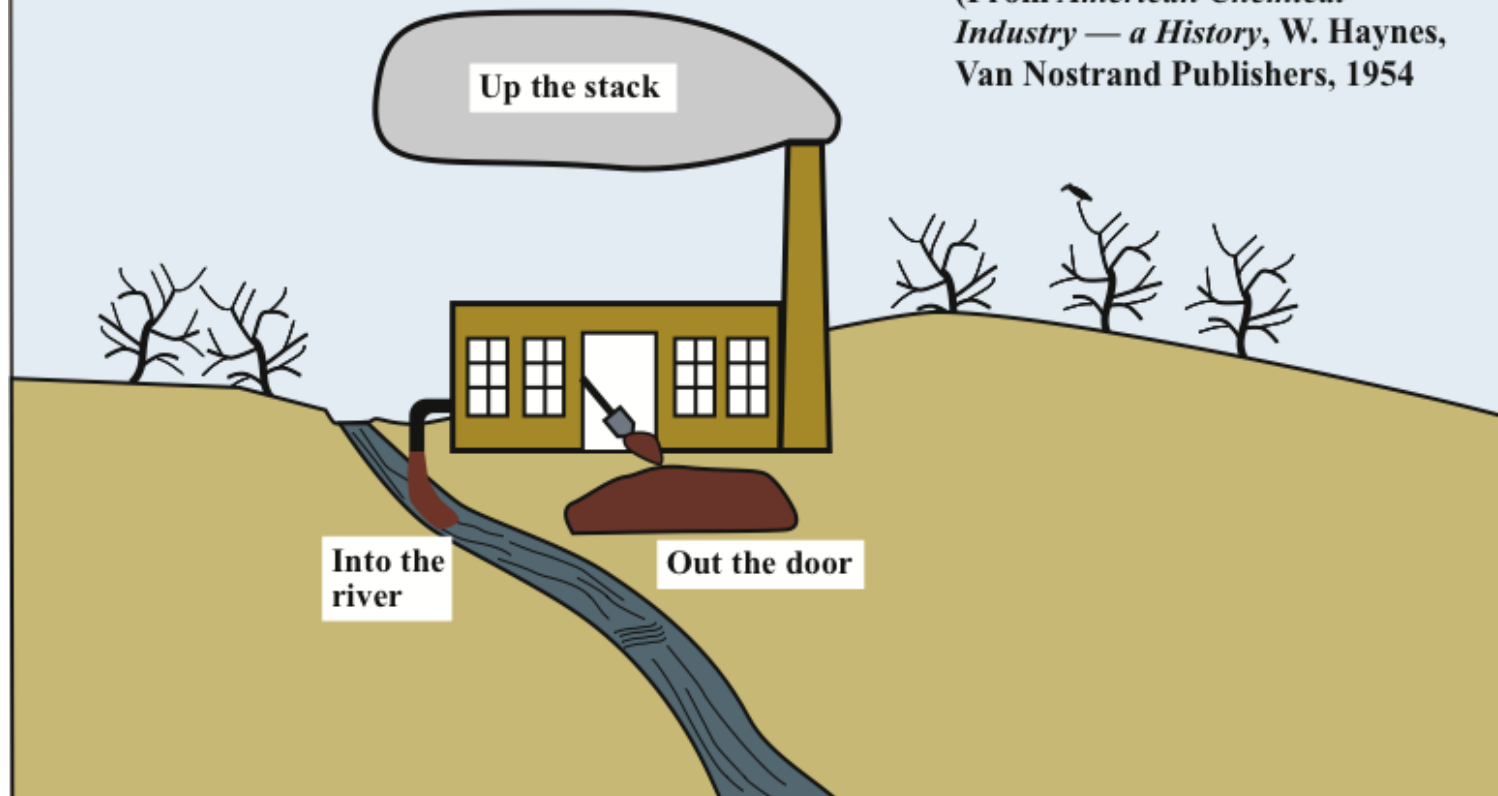
## The Carbon Cycle is Particularly Relevant to Climate Change



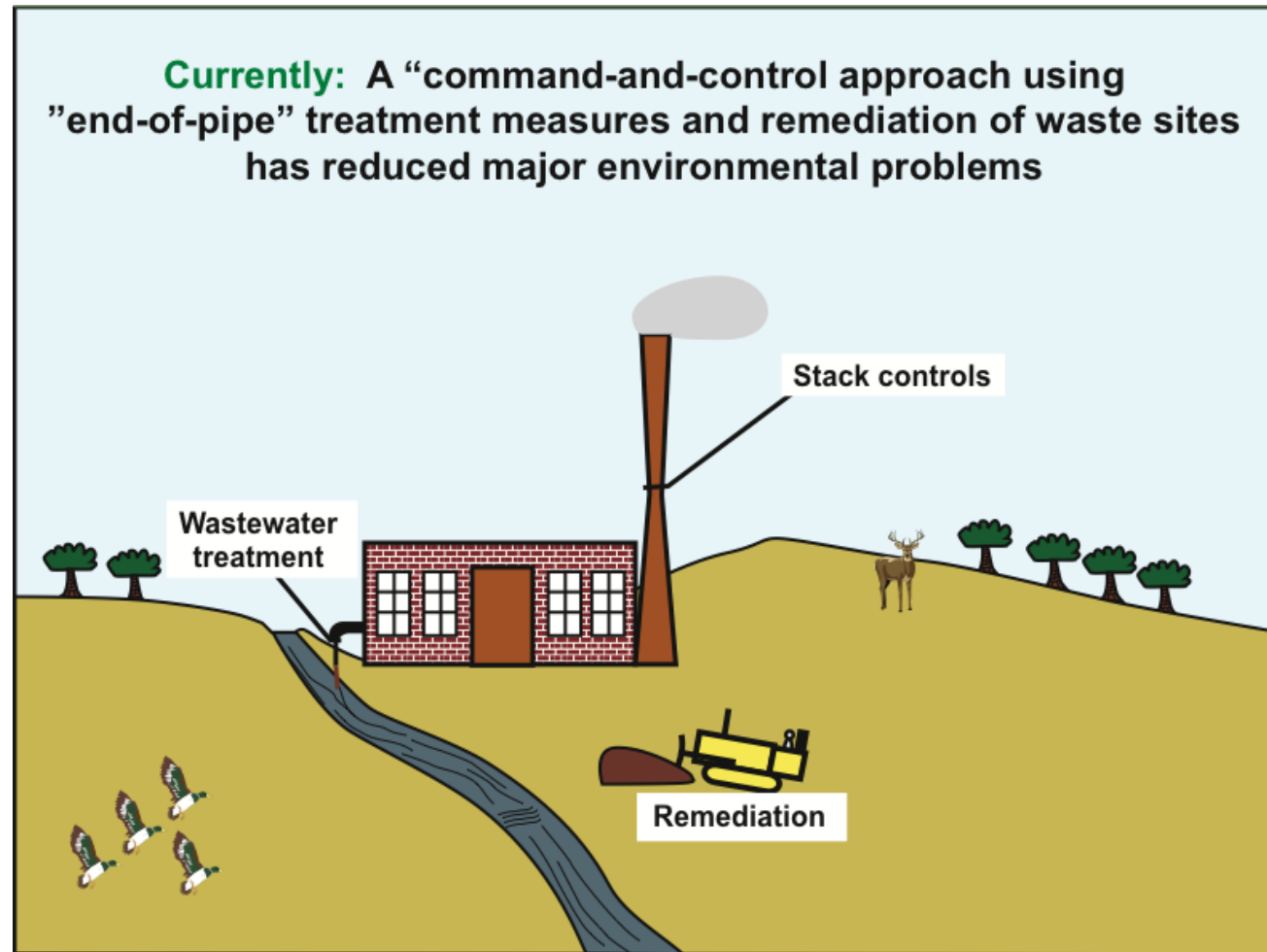
In the past many environmental problems were caused by practices that now seem to be totally unacceptable as expressed from the quote in this slide from what was regarded as a reputable book on the American chemical industry in 1954. The result was polluted air, polluted water, dangerous hazardous waste sites, and harm to living organisms.

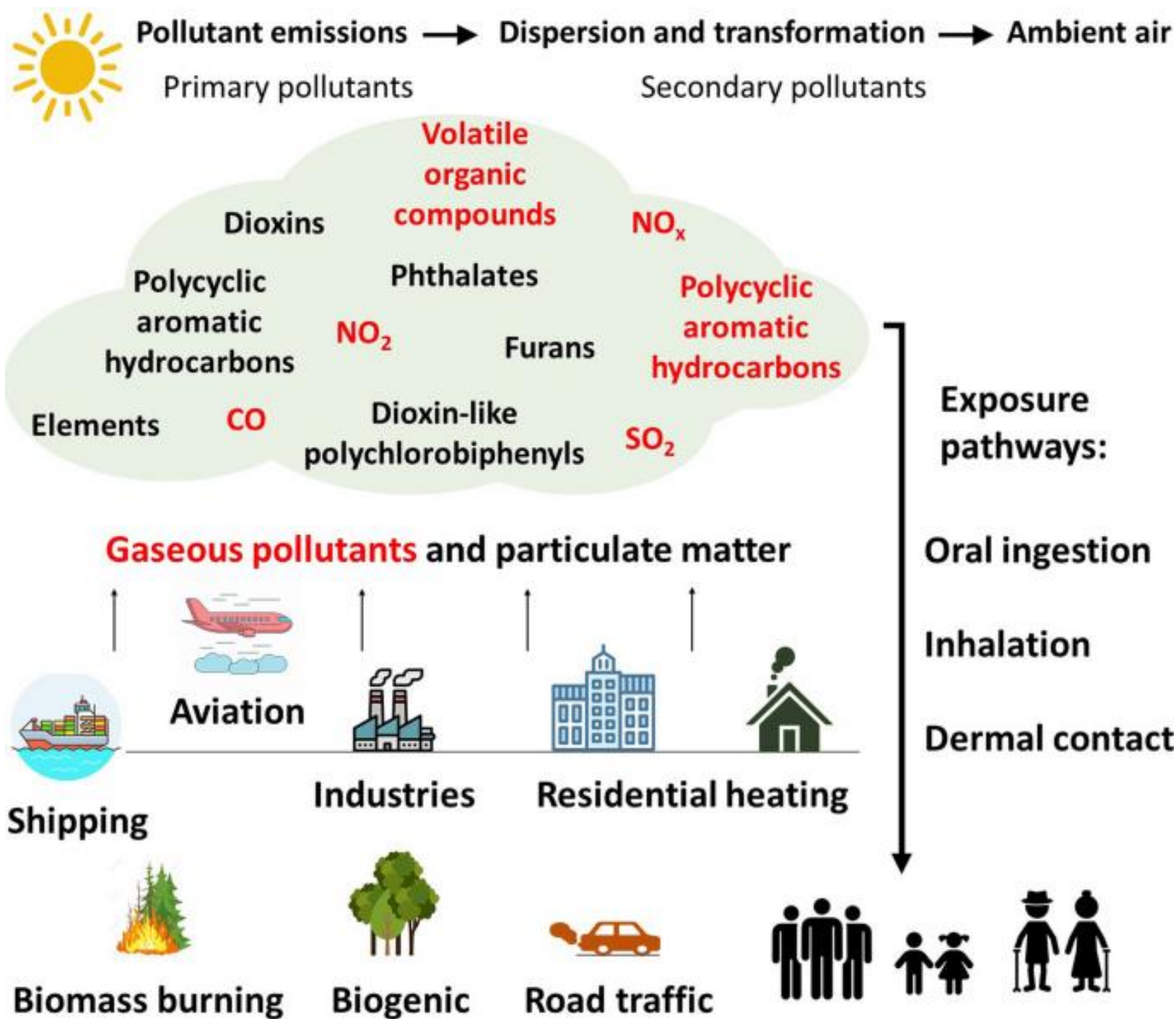
**The Old Attitude:** “By sensible definition any by-product of a chemical operation for which there is no profitable use is a waste. The most convenient, least expensive way of disposing of said waste — up the chimney or down the river — is best.”

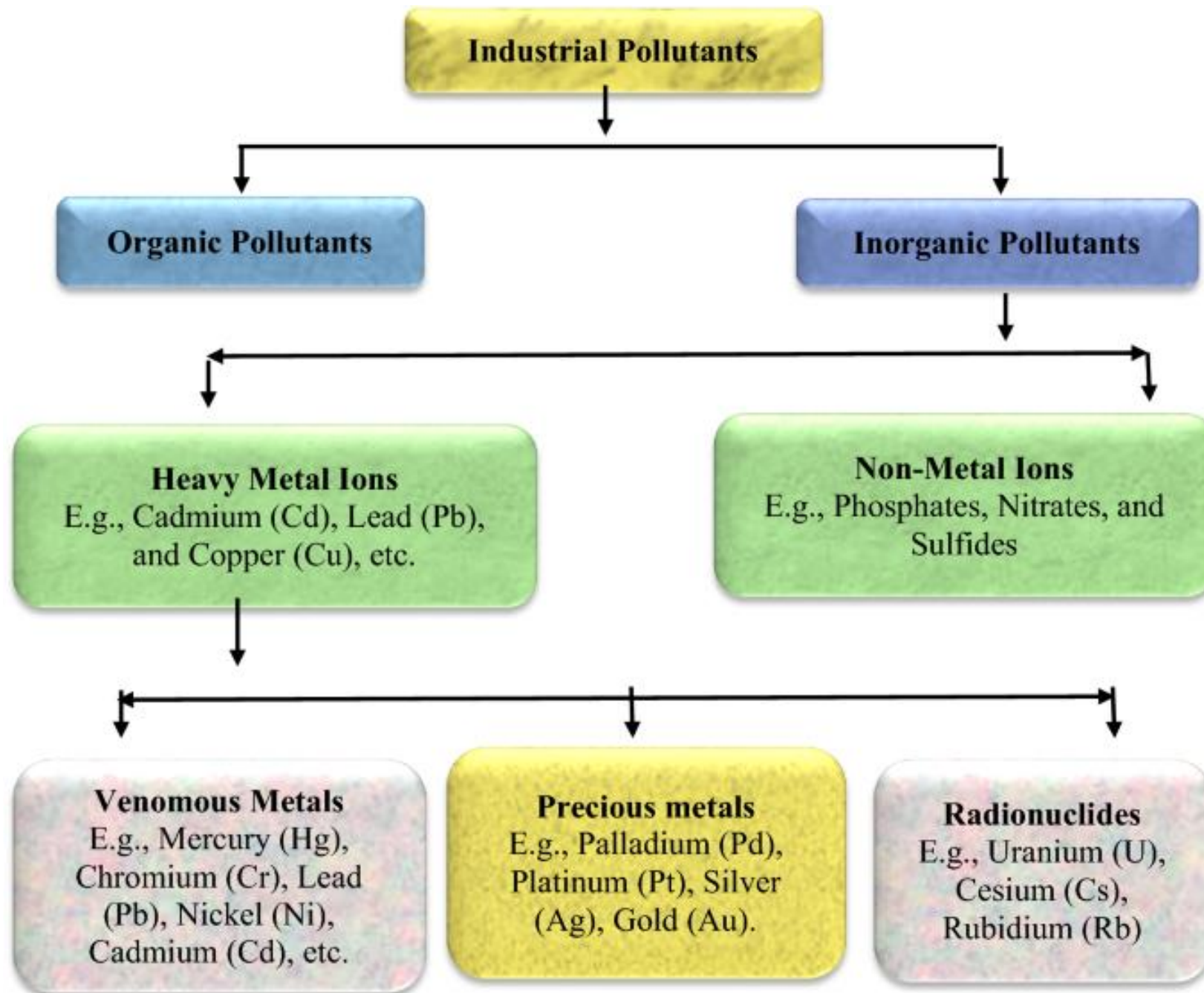
(From *American Chemical Industry — a History*, W. Haynes, Van Nostrand Publishers, 1954)

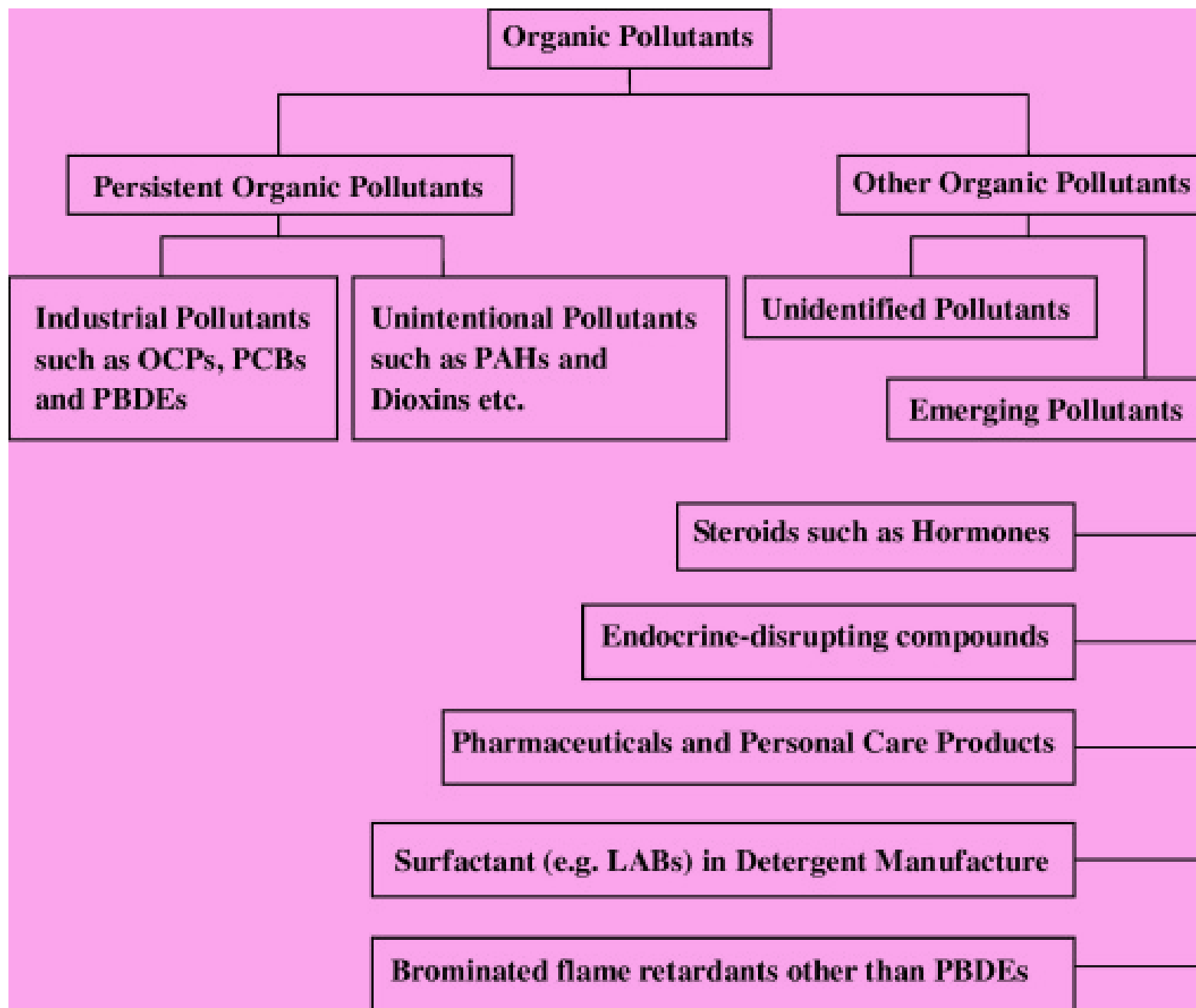


Dating from around 1970, laws and regulations were implemented to control air and water pollution and to clean up hazardous waste sites. These measures have relied largely upon “end of pipe” controls in which pollutants were generated but were removed before release to the environment. Although costly and requiring constant vigilance to make sure that standards have been met, these measures have been successful in reducing pollution and preventing increases in pollutant releases.







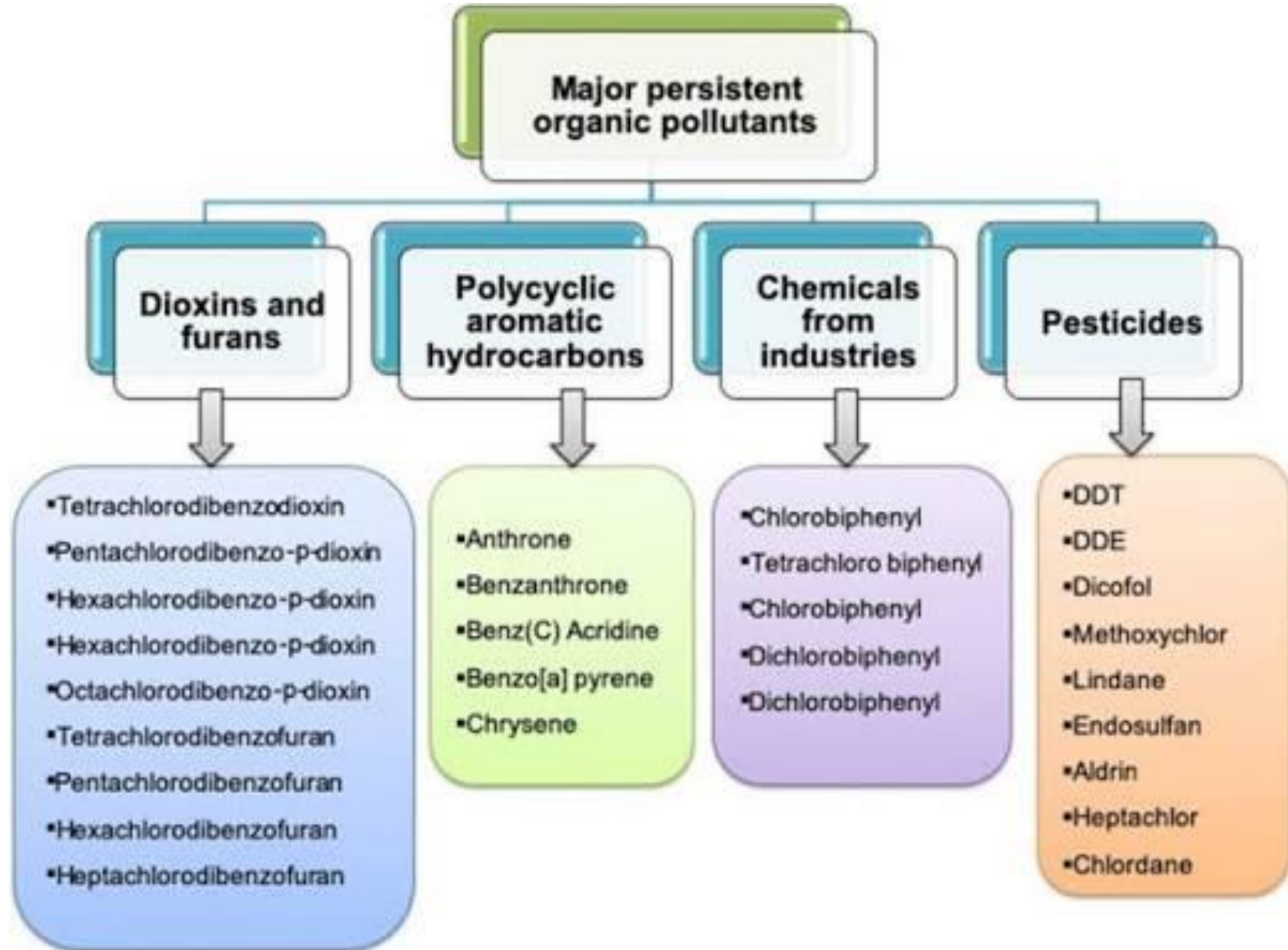


# Persistent Pollutants

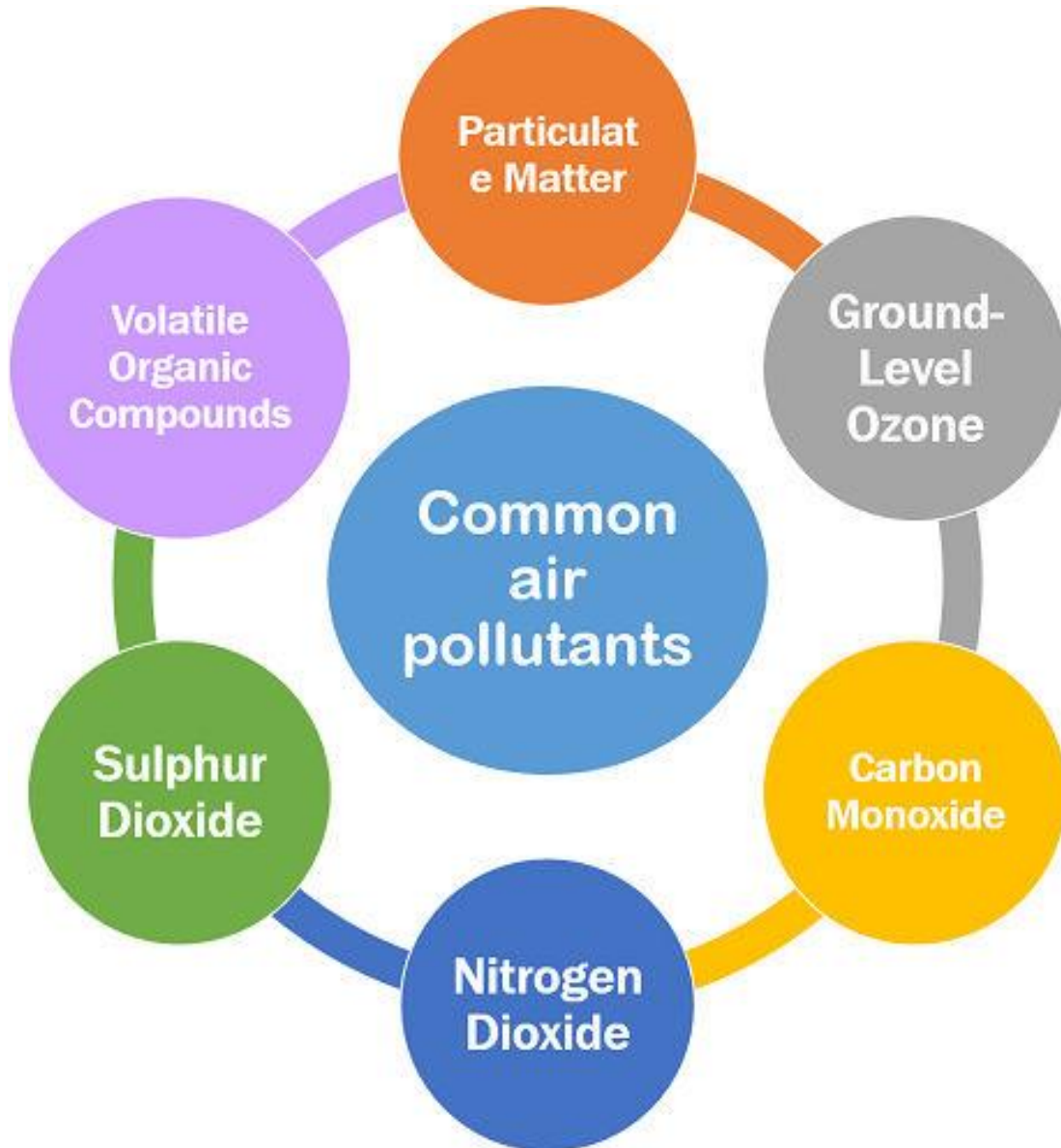
Toxic chemical substances that resist breakdown, remain in the environment for years, and accumulate in living organisms

## Key Characteristics

- **Long-lasting:** They do not easily break down via sunlight, water, or soil microbes.
- **Bioaccumulative:** They build up in the fatty tissues of animals and humans, growing stronger higher up the food chain (biomagnification).
- **Mobile:** Wind and water carry them across global borders far from their original source.



# Deeper analysis of atmospheric pollution



Air quality measurement are commonly reported in terms of:

- micrograms per cubic meter ( $\mu\text{g}/\text{m}^3$ )
- parts per million (ppm) or parts per billion (ppb)
- For particulate matter, sizes are expressed in micron or micrometer.

## Parts Per Million

$$\text{ppm} = \frac{\text{Mass of Solute (g)}}{\text{Mass of Solution (g)}} \times 10^6$$

# Analysis of air and air pollutants

- Analytical measurements form the foundation on which our understanding of the atmosphere is built
- In atmospheric chemistry, one of the most fundamental measurements is the most difficult.
- What is the composition of air?
- Nitrogen gas represents more than 78% of the atmosphere, whereas reactive species such as ozone are present in ppm level. The range in concentration of atmospheric constituents is extremely wide, as large as  $10^4$

# Analysis of air and air pollutants

## Two different measurement

### Techniques:

***In situ:*** It means the sample of the atmospheric gases must be placed inside the device (spectrometer) that is making

***Remotely:*** (Remote sensing Technique): In this case by passing a beam of energy that originates on a satellite, an aircraft, the space shuttle or the ground through a portion of the atmosphere that is to be studied.

***Ex-situ:*** *Collect the sample and analyse it in the laboratories*

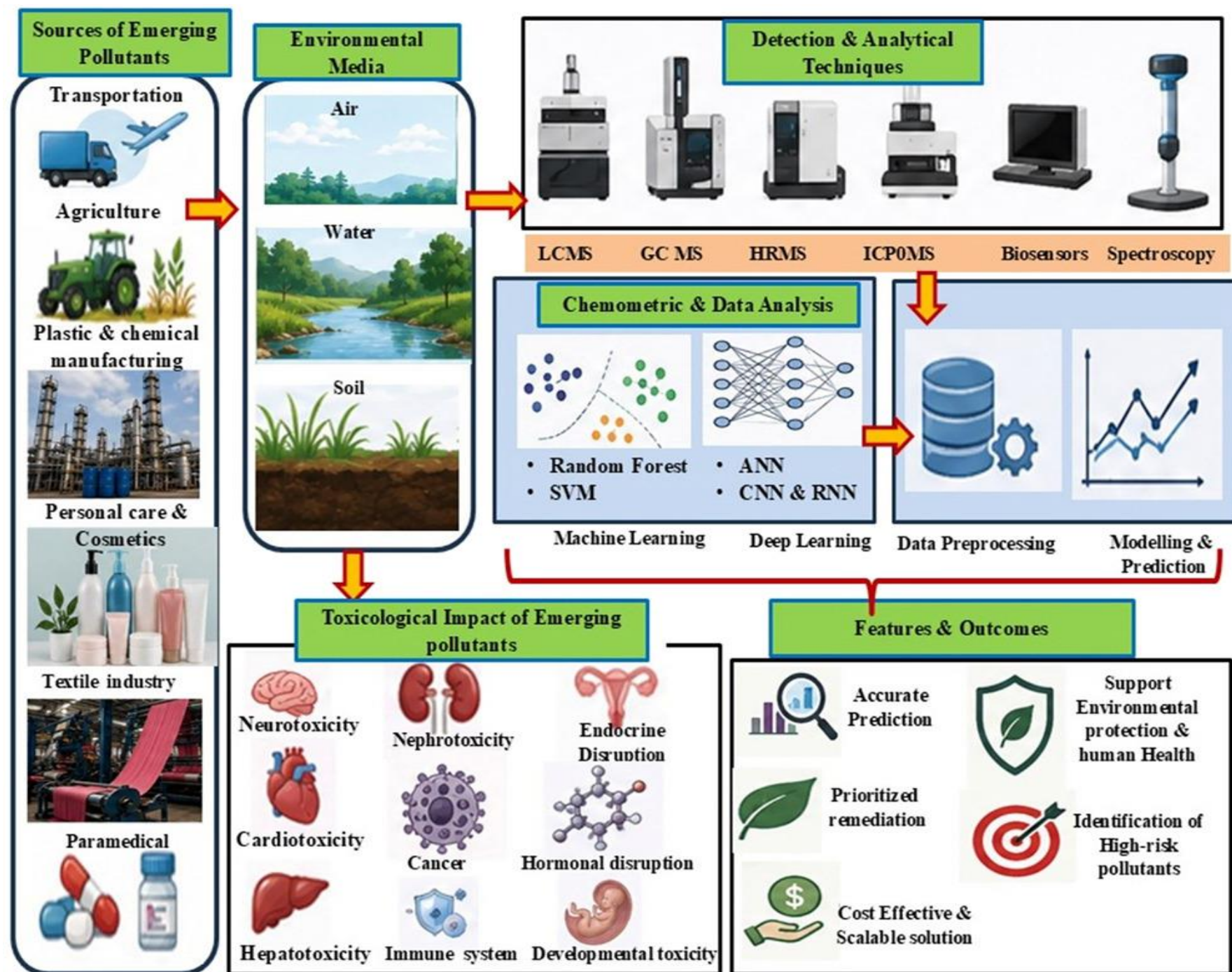
Although these measurement techniques involve different instrumentation, the underlying principles are same.

# Analysis of air and air pollutants

- Automobile and smoke stack emissions are major sources of **anthropogenic pollutants** into the atmosphere, both of which are regulated by the Environmental Protection Agency (EPA).

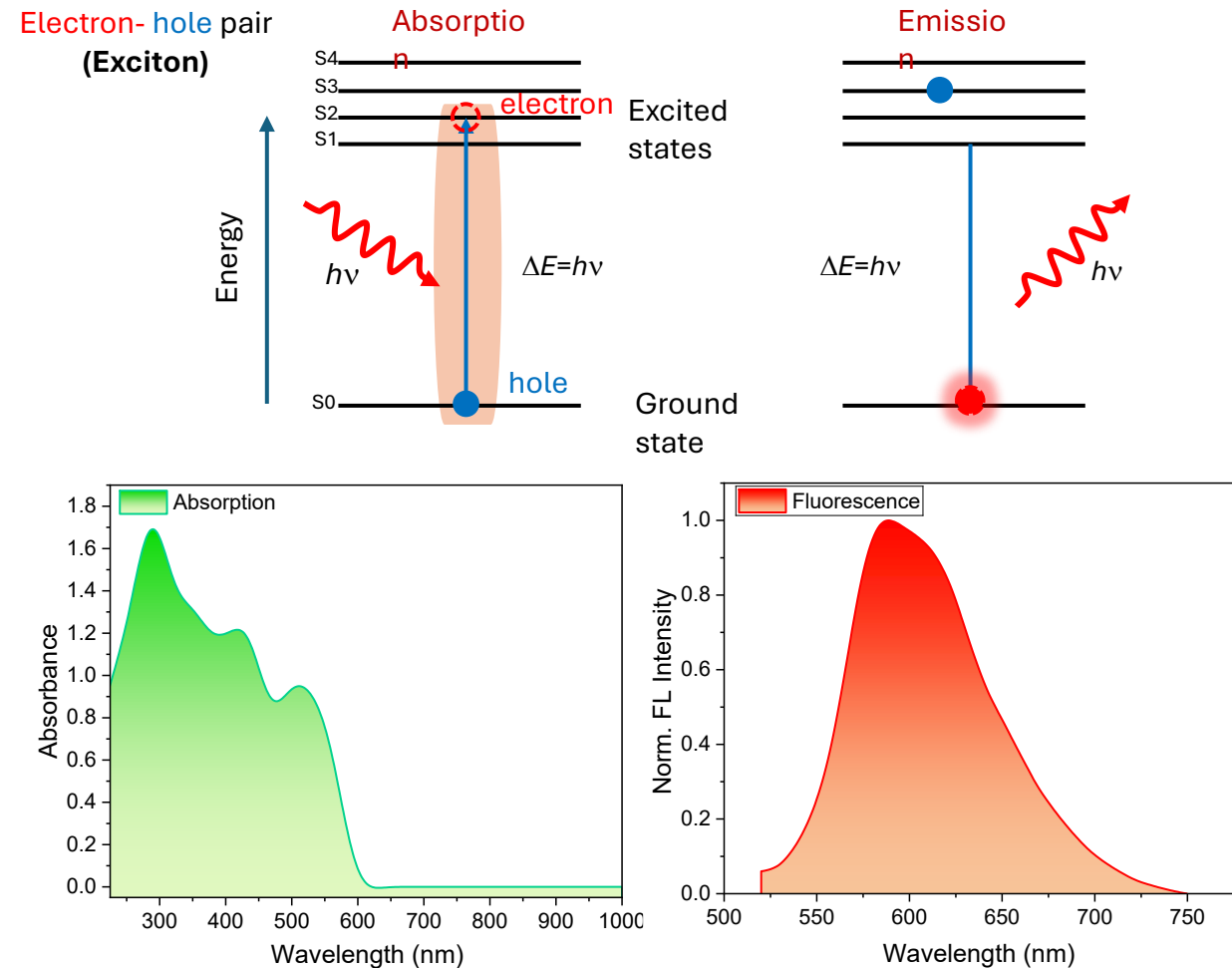
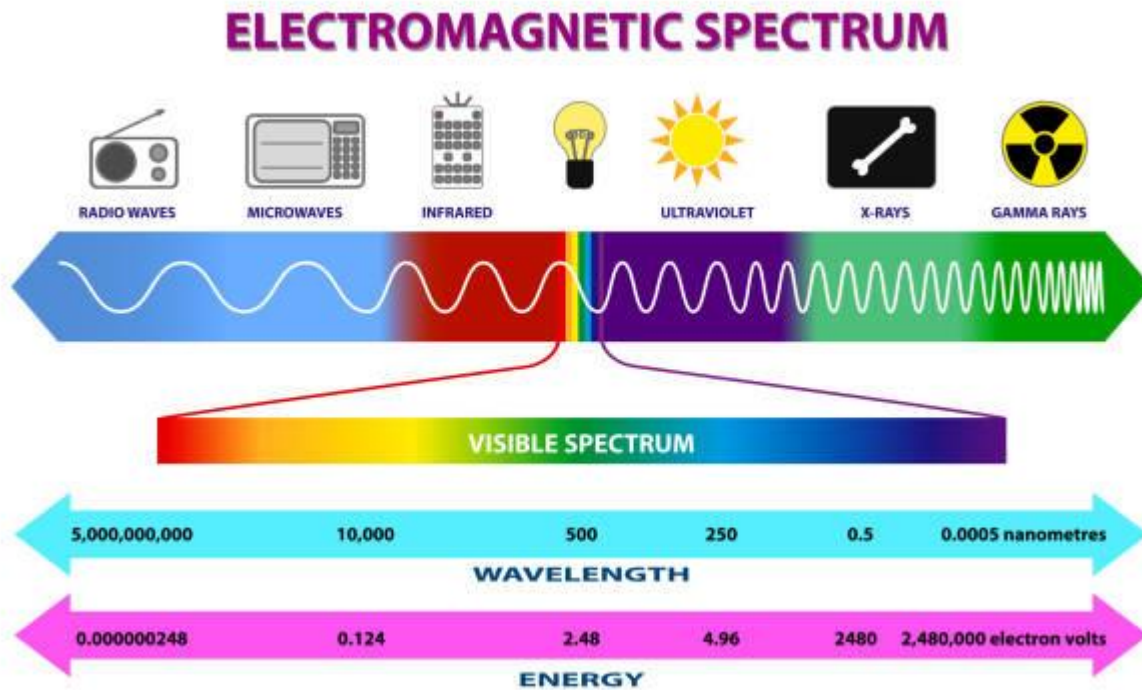


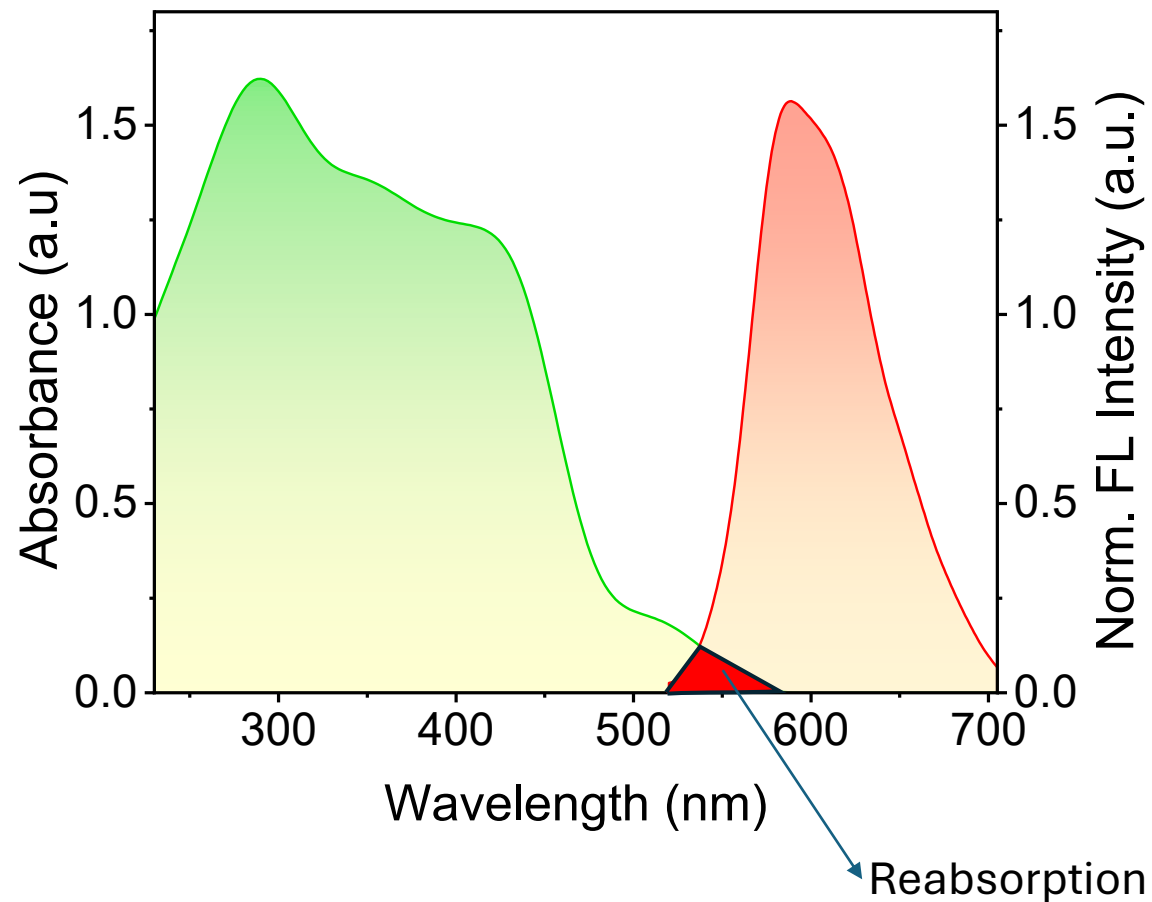
- Measurements of these gases emitted from automobiles and smoke stacks are difficult to make because the exhaust gases are hot and travel at a high velocity.



# In situ Absorption Measurement

- Spectroscopic Measurement:** This method relies on the fact that different chemical compounds absorb electromagnetic radiation at different wavelengths. When a molecule absorbs a photon of electromagnetic radiation, the energy of the molecule increases.





|            |                              |                 |
|------------|------------------------------|-----------------|
| $A$        | Absorbance                   |                 |
| $\epsilon$ | Molar absorption coefficient | $M^{-1}cm^{-1}$ |
| $c$        | Molar concentration          | $M$             |
| $l$        | optical path length          | $cm$            |

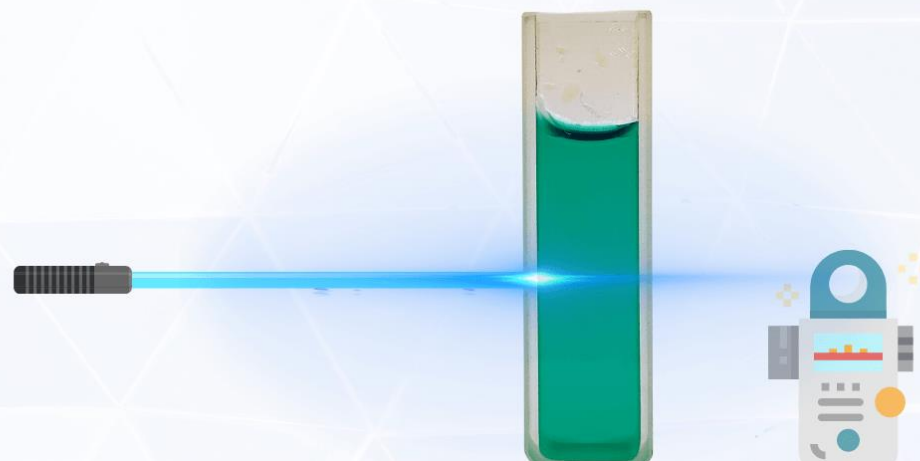
$$A = \epsilon cl$$

## Beer's Law

Light absorption is directly proportional to path length and the concentration of the solution.

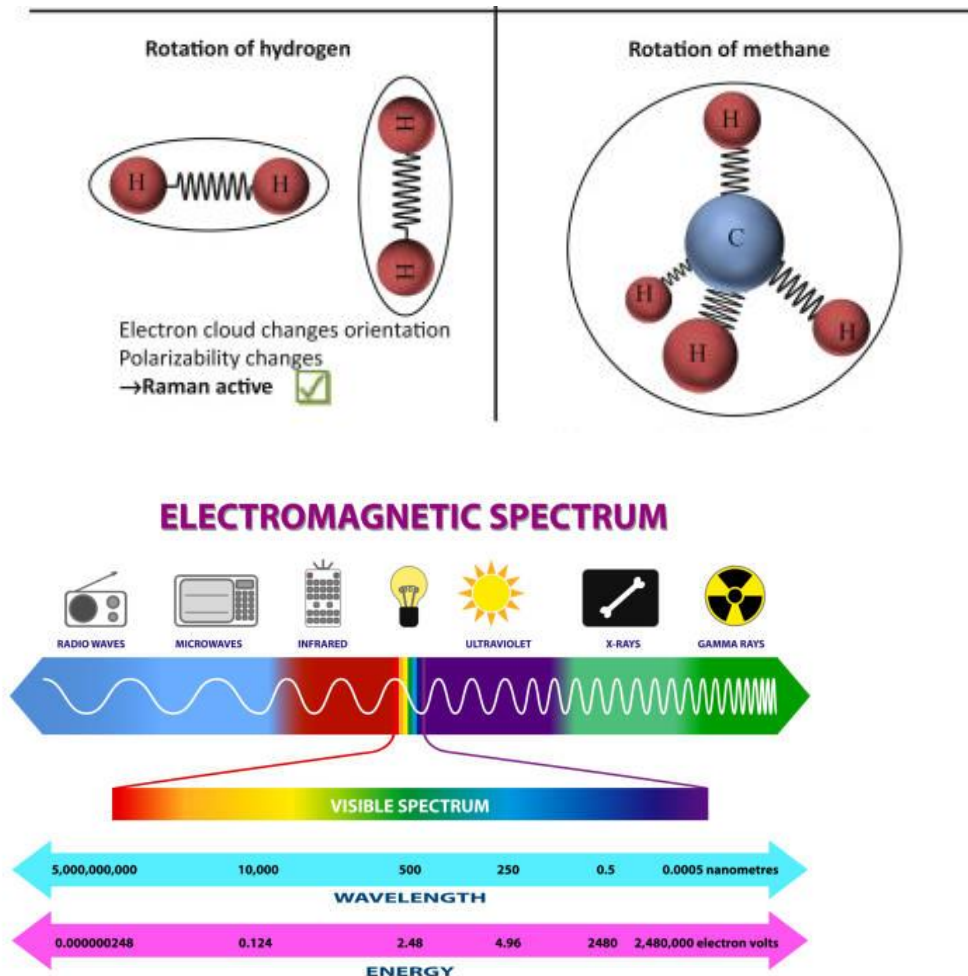
$$A = \epsilon l c$$

absorbance      molar absorptivity      path length      concentration

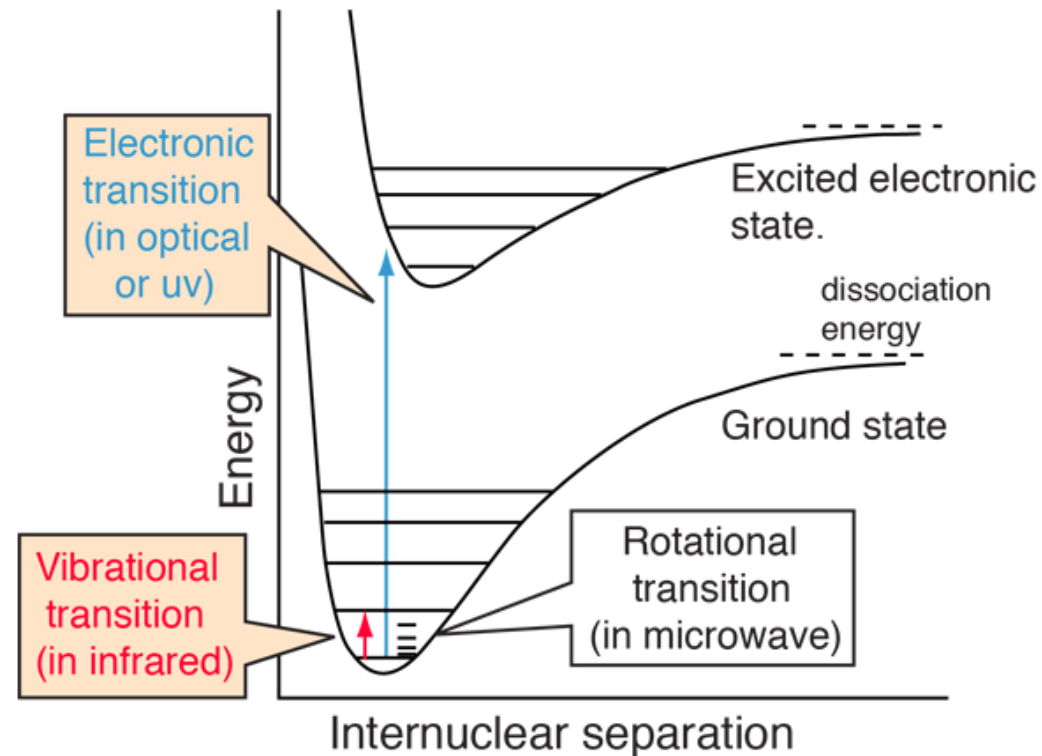


# In situ Absorption Measurement

- For example, when a molecule absorbs microwave radiation, which is EMR of relatively low energy, it only stimulates the rotational motion of the molecule.

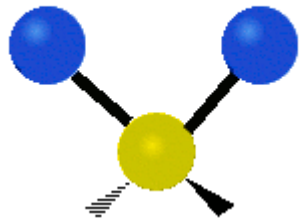


|                  |            |                      |                             |
|------------------|------------|----------------------|-----------------------------|
| <b>Microwave</b> | 1 mm – 1 m | 300 GHz –<br>300 MHz | 1.24 meV –<br>1.24 $\mu$ eV |
|------------------|------------|----------------------|-----------------------------|

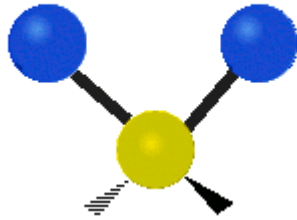


# In situ Absorption Measurement

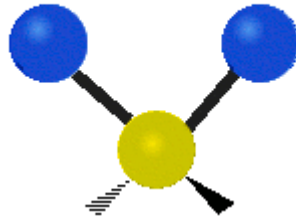
- IR radiation which has higher energy than microwaves, **stimulates vibrations of the molecules that absorb it.** UV radiation which has even higher energy than IR, causes electrons in the molecules absorbing it to be promoted into higher energy orbitals; the molecule is said to be in the excited state. The lowest energy state of the molecule is called the ground state.



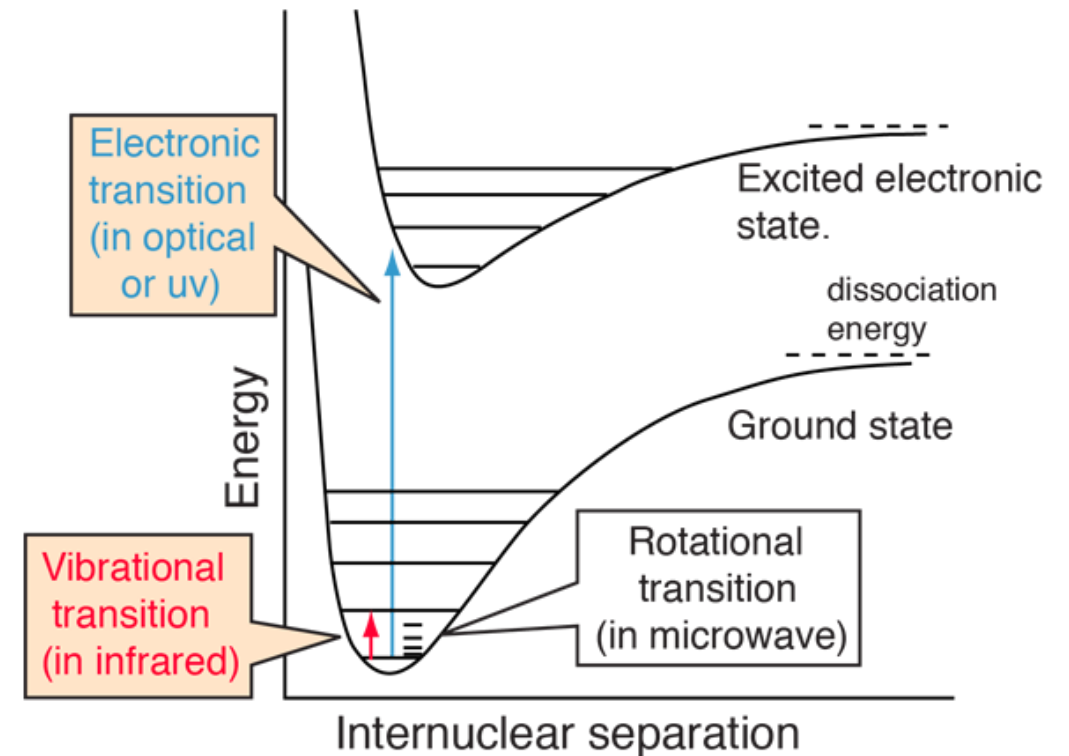
Symmetric Stretch



Asymmetric Stretch



Twisting



|                 |                  |                      |                      |
|-----------------|------------------|----------------------|----------------------|
| <b>Infrared</b> | 700 nm –<br>1 mm | 300 GHz –<br>430 THz | 1.7 eV –<br>1.24 meV |
|-----------------|------------------|----------------------|----------------------|

# In situ Absorption Measurement

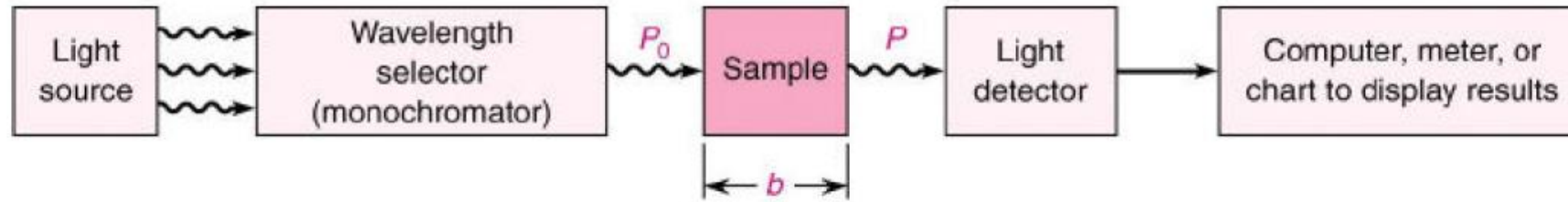
- Very high energy electro magnetic radiation (EMR) such as X-ray and  $\gamma$  has enough energy to break chemical bonds and ionize molecules.



- When the sample absorbs a beam of EMR radiation, the irradiance of the beam is decreased. The irradiance ( $P$ ), which is sometimes intensity or radiant power, is the energy per second per unit area of the light beam.
- EMR is passed through a monochromator (a prism) to select one wavelength of EMR. The light of a single wavelength is called monochromatic, which means one color.
- The monochromatic light with irradiance  $P_0$  passes into a sample  $b$ . The irradiance of the beam emerging from the other side of the sample is  $P$ . Some of the light may be absorbed by the sample, and

# In situ Absorption Measurement

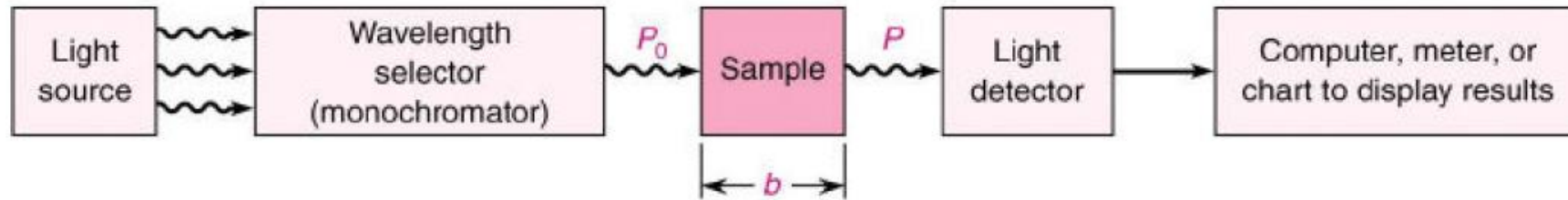
## *Instrumentation*



- **Transmittance (T):** it is defined as the fraction of the original light that has passed through the sample.
- $T = \frac{P}{P_0}$  T has the range of 0 to 1. The percentage of transmittance is  $100 T$  and has a range from 0% to 100%. Absorbance (A) is defined as follows:
- $A = \log \left( \frac{P_0}{P} \right) = -\log T$  ; When no light is absorbed  $P = P_0$  and  $A = 0$

# In situ Absorption Measurement

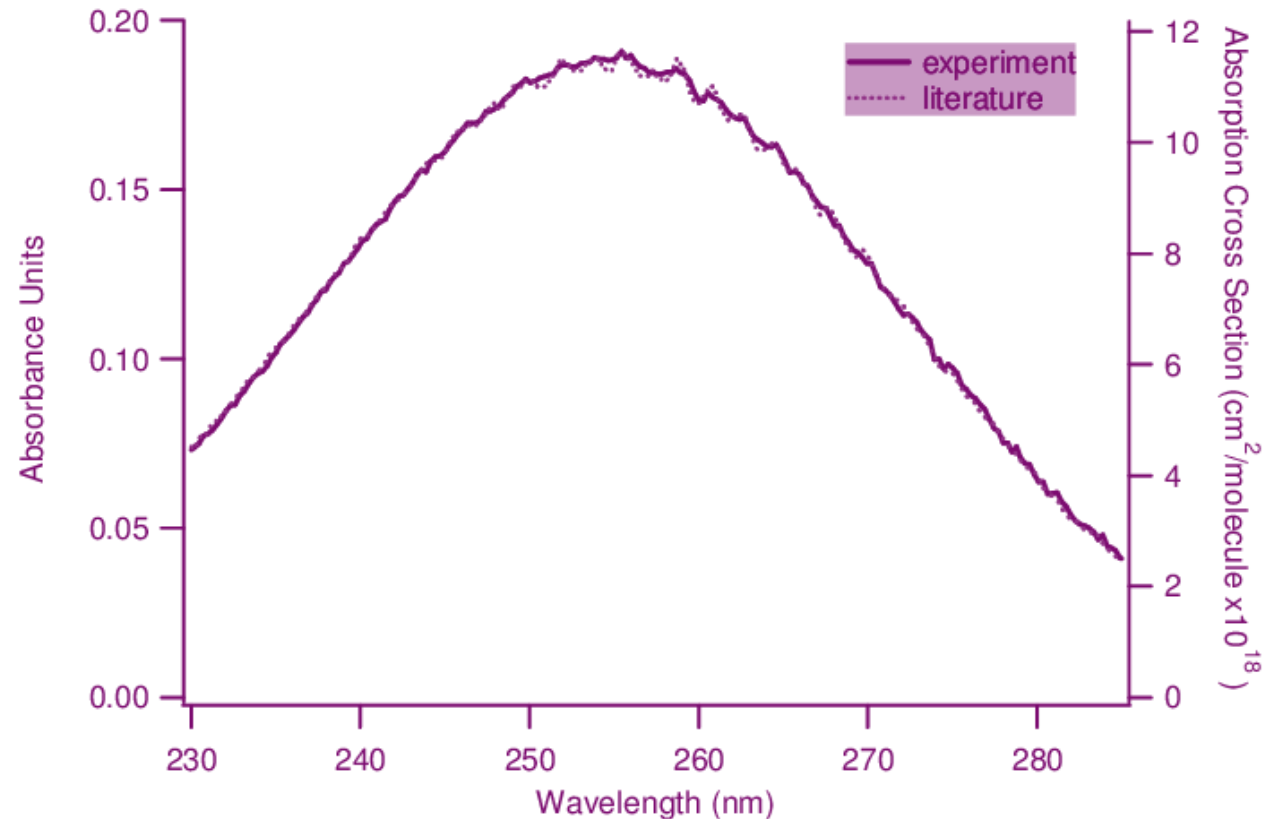
## Instrumentation



- Absorbance is important because it is directly proportional to the concentration (C) of the absorbing molecules in the sample. **A is dimensionless.**
- The Beer-Lambert or simply Beer's Law: 
$$A = \epsilon C l$$
- $A = \epsilon \cdot b \cdot C$  (b = path length cm or m etc., C = concentration moles/L, ppm, ppb etc.,)
- $\epsilon$  = molar absorptivity or extinction coefficient. It is the characteristics of the molecule that indicates how much light it will absorb at a particular wavelength.

# In situ Ozone measurement

- Ultraviolet* absorption has been used to measure the concentration of ozone in stratospheric air. The figure shows that ozone has a strong absorption in the UV region. Other atmospheric gases such as O<sub>2</sub>, N<sub>2</sub>, and H<sub>2</sub>O do not absorb UV radiation and thus do not have to be removed from the air sample as they do not interfere with the ozone measurement.



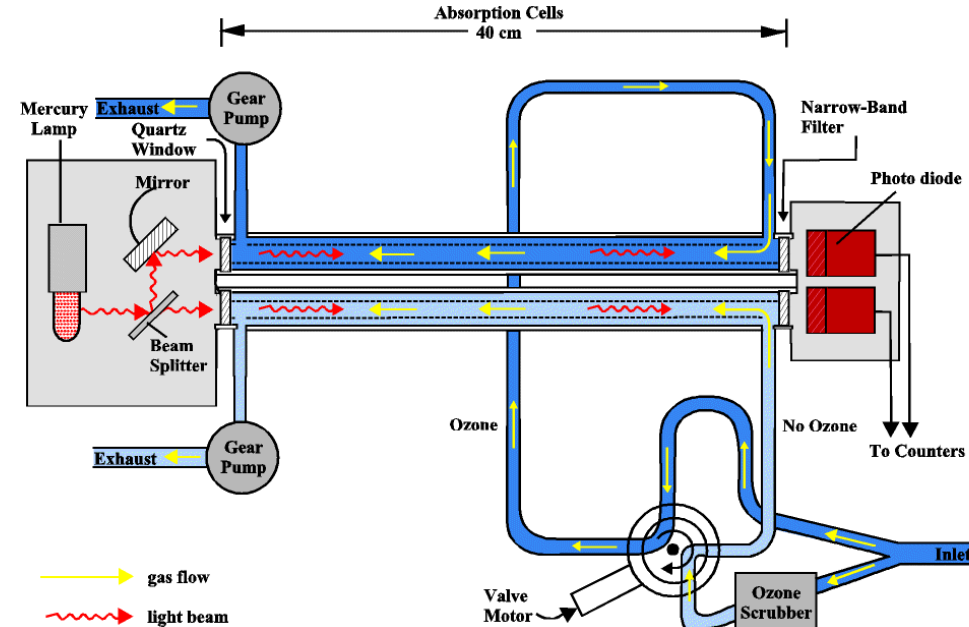
# In situ Ozone measurement

- The concentration of ozone in the stratosphere is less than 100 ppm, so the spectrophotometer that is used to make the measurement must be capable of measuring a small absorbance. This is most easily done by making the sample cell very long. Because Beer's law says that absorbance is proportional to the sample path length and because we can get a lot of samples easily, a long sample path length will greatly improve our ability to measure the ppm level of Ozone.

## Ozone: Dual Beam UV-Photometer

[Proffitt, M.H., and R.J. McLaughlin, Rev. Sci. Instrum., 54, 1719-1728, 1983]

[Proffitt, M.H., and R.J. McLaughlin, Rev. Sci. Instrum., 54, 1719-1728, 1983]



# Basic Pollutants

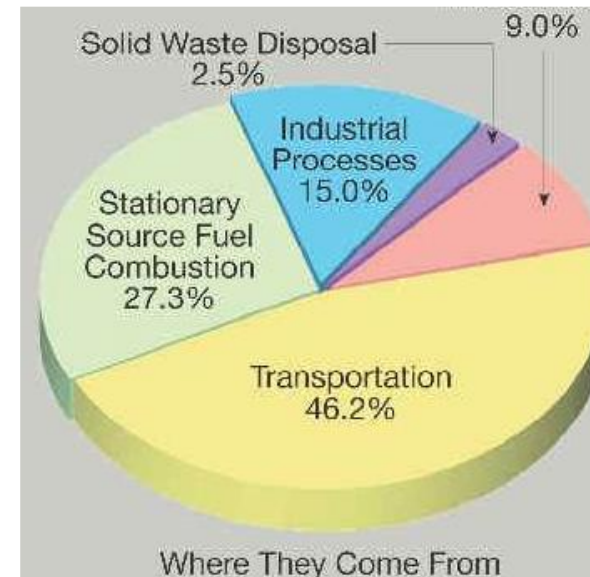
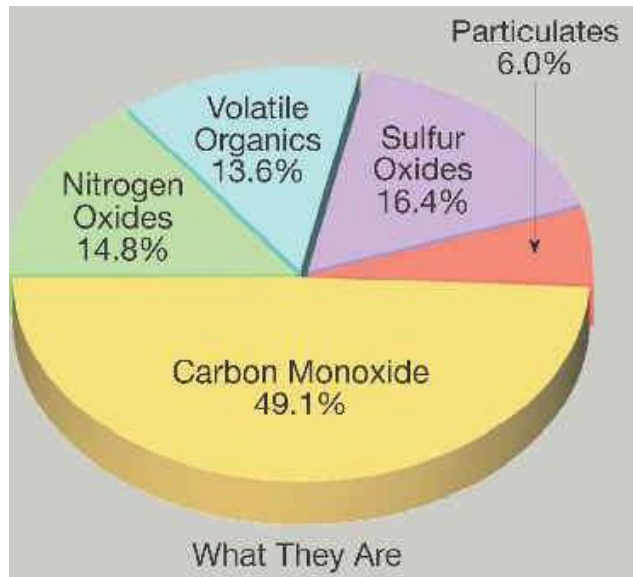
## Basic Pollutants

| <u>Pollutant</u>                                                       | <u>Abbreviation</u> | <u>Type</u>         |
|------------------------------------------------------------------------|---------------------|---------------------|
| Carbon Monoxide                                                        | CO                  | Primary             |
| Sulfur Dioxide                                                         | SO <sub>2</sub>     | Primary             |
| Ozone                                                                  | O <sub>3</sub>      | Secondary           |
| Nitrogen Dioxide                                                       | NO <sub>2</sub>     | Secondary           |
| Hydrocarbon Compounds (also called VOCs – volatile organic compounds ) | HC                  | Primary & Secondary |
| Particulate Matter                                                     | PM                  | Primary & Secondary |
| Lead                                                                   | Pb                  | Primary & Secondary |

# Types of Pollutants

**Primary pollutants** are those gases or particles that are pumped into the air to make it unclean. They include carbon monoxide from automobile (cars) exhausts and sulfur dioxide from the combustion of coal. Primarily air pollutants can be caused by primary sources or secondary sources. **The pollutants that are a direct result of the process can be called primary pollutants.**

A classic example of a primary pollutant would be the sulfur-dioxide emitted from factories.



# Chemistry of Troposphere

## ❖ Air Pollutant

➤ An air pollutant is defined as a substance that is present in the atmosphere at a concentration that is sufficient to cause harm to humans, other animals, vegetation or materials

➤ **Approx 90% of all air pollution is caused by these primary air pollutants**

1. Carbon monoxide (CO)
2. Sulfur dioxide (SO<sub>2</sub>)
3. Nitrogen Oxides (NO<sub>x</sub>)
4. Volatile Organic Compounds (VOCs)
5. Mostly Hydrocarbons (HCs)
6. Suspended particles

➤ **In addition, Secondary pollutants are also present**

➤ Harmful substances are produced by chemical reactions between primary pollutants and other constituents of the atmosphere. e.g., H<sub>2</sub>SO<sub>4</sub>, HNO<sub>3</sub>; SO<sub>4</sub><sup>2-</sup> and NO<sub>3</sub><sup>-</sup> contri



**Ozone and other photochemical oxidants contribute to photochemical smog**

# Chemistry of Troposphere

## ❖ Carbon Monoxide

### ➤ Source of Carbon monoxide

➤ The main anthropogenic source of CO is the combustion of gasoline in automobile engines

1. Gasoline is a complex mixture of hydrocarbons (HCs)
2. If this is ignited in an adequate supply of oxygen; the products are CO<sub>2</sub> and H<sub>2</sub>O

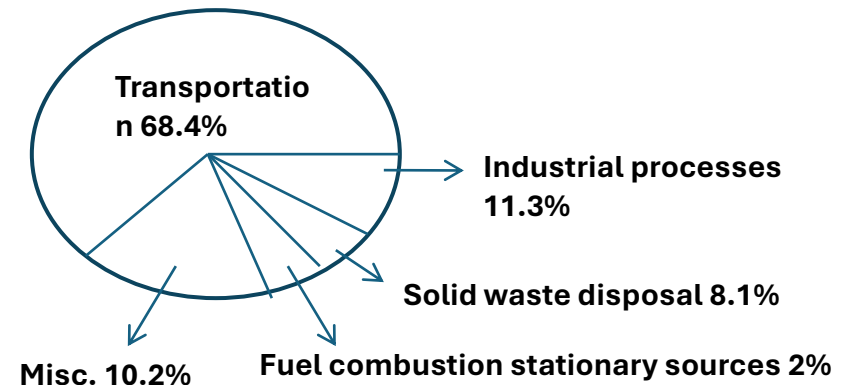


1. In the confined space of the internal combustion of the engine O<sub>2</sub> supply is limited, and combustion is incomplete. Thus, CO is formed and released into the atmosphere in automobile exhaust



❖ However, the introduction of catalyst converter  
reduce the CO emissions

❖ Natural source of CO into the atmosphere is CH<sub>4</sub>  
gas



Methane is also produced in the stomach in the cattle and sheep and finally released into the atmosphere

# Carbon Monoxide

## ❖ Effect of CO on Human Health

- Even the CO is most abundant air pollutant, it is not very toxic at the levels usually found in the atmosphere, however if allowed a build up in a confined space, it can cause serious health problem
- CO interferes with the oxygen carrying capacity of blood; normally hemoglobin (Hb) in red blood cells combine with oxygen in the lungs to oxyhemoglobin (HbO<sub>2</sub>). The HbO<sub>2</sub> is carried in the blood stream to various part of the body, where oxygen is released to the tissues
- Carbon monoxide binds much more strongly to Hb than oxygen – Why?
- If CO is present in the lungs, it displaces oxygen from Hb and thus reduces the amount of oxygen that can be delivered to the tissues



## ❖ Treatment of CO poisoning:

- ❖ Symptoms of CO poisoning: headache, dizziness, impaired judgment, drowsiness, slowed reflexes, respiratory failure and death

# Symptoms of Carbon Monoxide Poisoning

## Mild Exposure



Headache



Dizziness



Nausea



Vomiting



Fatigue



Confusion

## Moderate Exposure



Weakness



Shortness  
of Breath



Blurred  
Vision



Difficulty  
Concentrating

## Severe Exposure



Chest  
Pain



Loss of  
Consciousness



Seizures



Death

# Carbon monoxide poisoning

## Symptoms of Carbon monoxide poisoning

- Dizziness
- Headache
- Disorientation
- Impairment of the cerebral function
- Coma

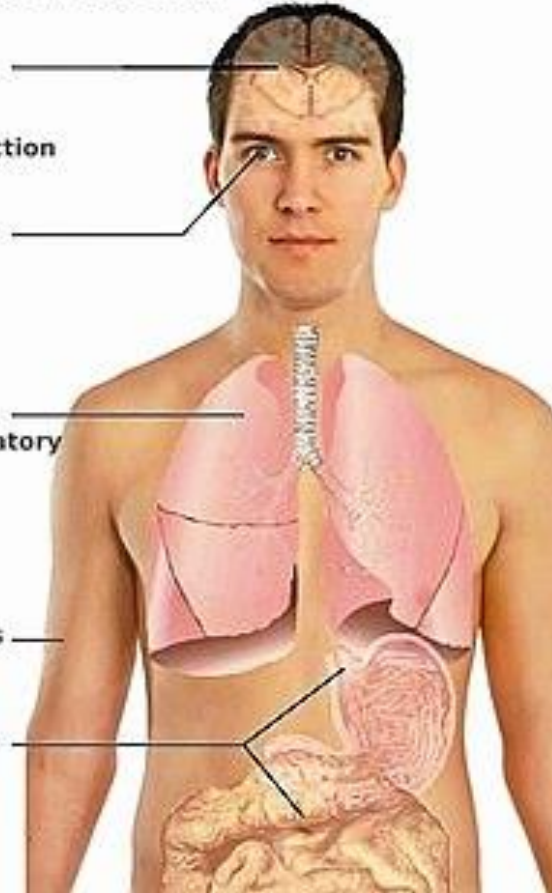
- Visual disturbances

- Disease of the heart and respiratory

- Muscle weakness
- Muscle cramps
- Seizures

- Nausea

- Aggravation of preexisting diseases



35 ppm (0.0035%)

Headache &  
Dizziness

200 ppm (0.02%)

Headache &  
loss of  
judgement

3200 ppm (0.32%)

Headache,  
Dizziness,  
Nausea

6400 ppm (0.64%)

Respiratory  
arrest & death  
in less than 20  
min

12800 ppm (1.28%)

Unconsciousne  
ss. Death in less  
than 3 min